Perceptual Organization (IV)

Introduction to Computational and Biological Vision

CS 202-1-5261

Computer Science Department, BGU

Ohad Ben-Shahar

Segmentation

Segmentation as partitioning

Given: ***I*** **- a set of image pixels**

H **– a region homogeneity predicate (based on visual properties of interest)**

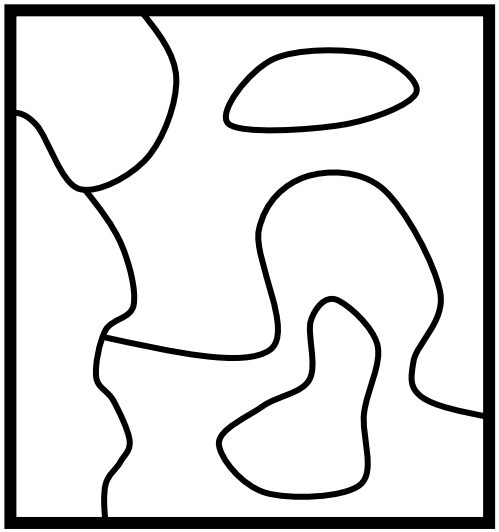
a segmentation of the image is a (meaningful) partition of ***I*** **into** ***m*** **regions** R_i **such that**

$$R_i \cap R_j = \varnothing \qquad \forall i \neq j$$

$$\bigcup_{i=1}^{m} R_i = I$$

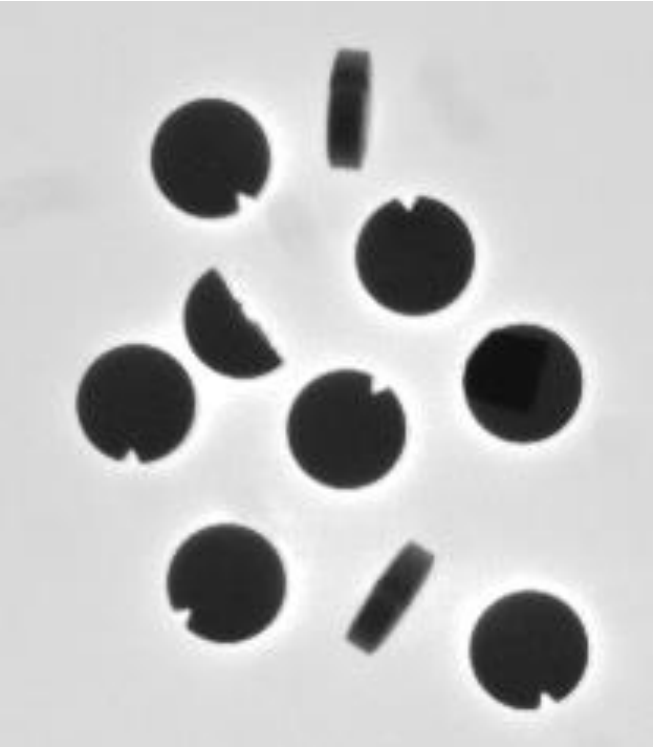
$$H(R_i) = true \qquad \forall i$$

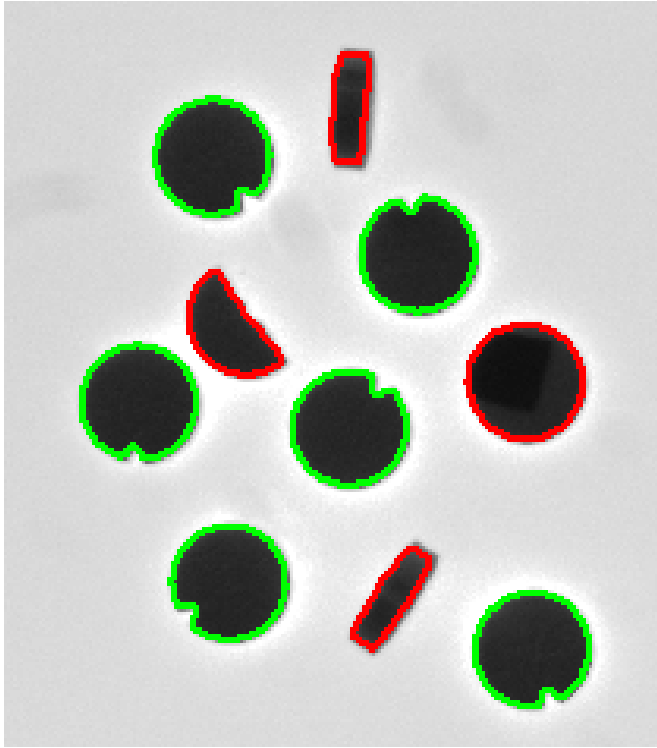
$$H(R_i \cup R_i) = false \qquad \forall i \neq j \quad \text{adjacent}$$



Segmentation

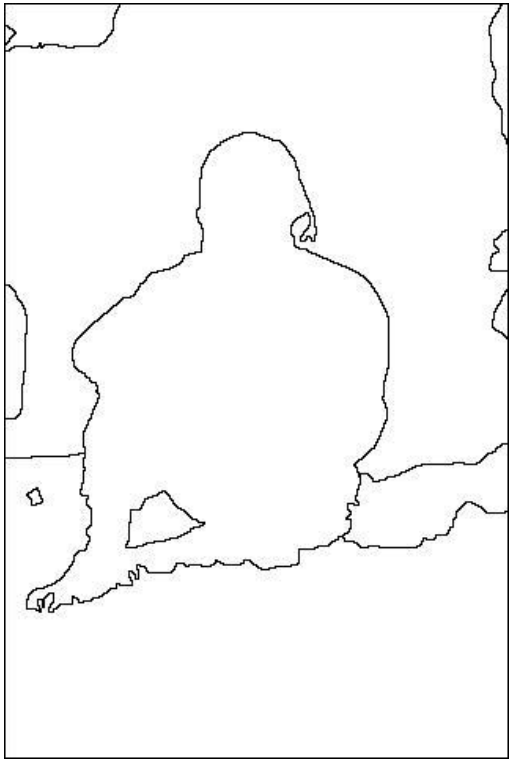
Is segmentation easy?





Segmentation

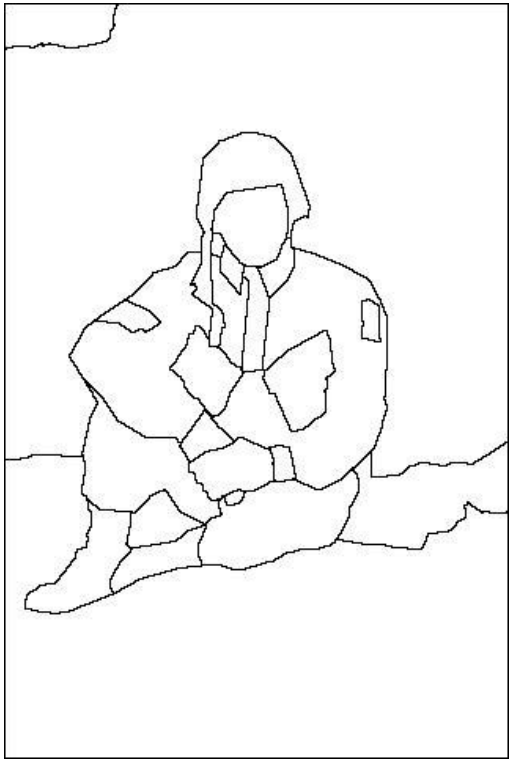
Segmentation is conceptually ill defined



12 segments

Segmentation

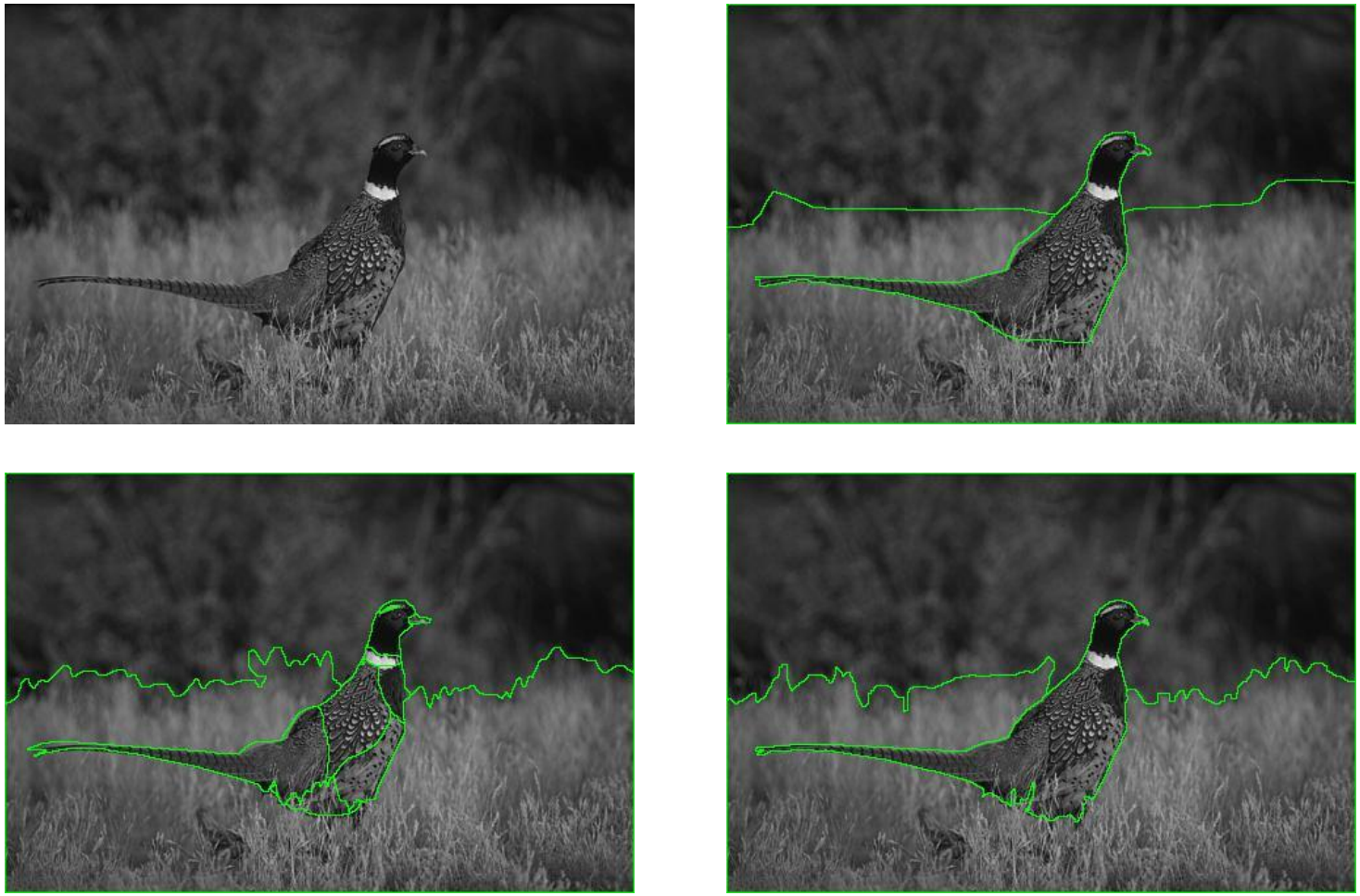
Segmentation is conceptually ill defined



25 segments

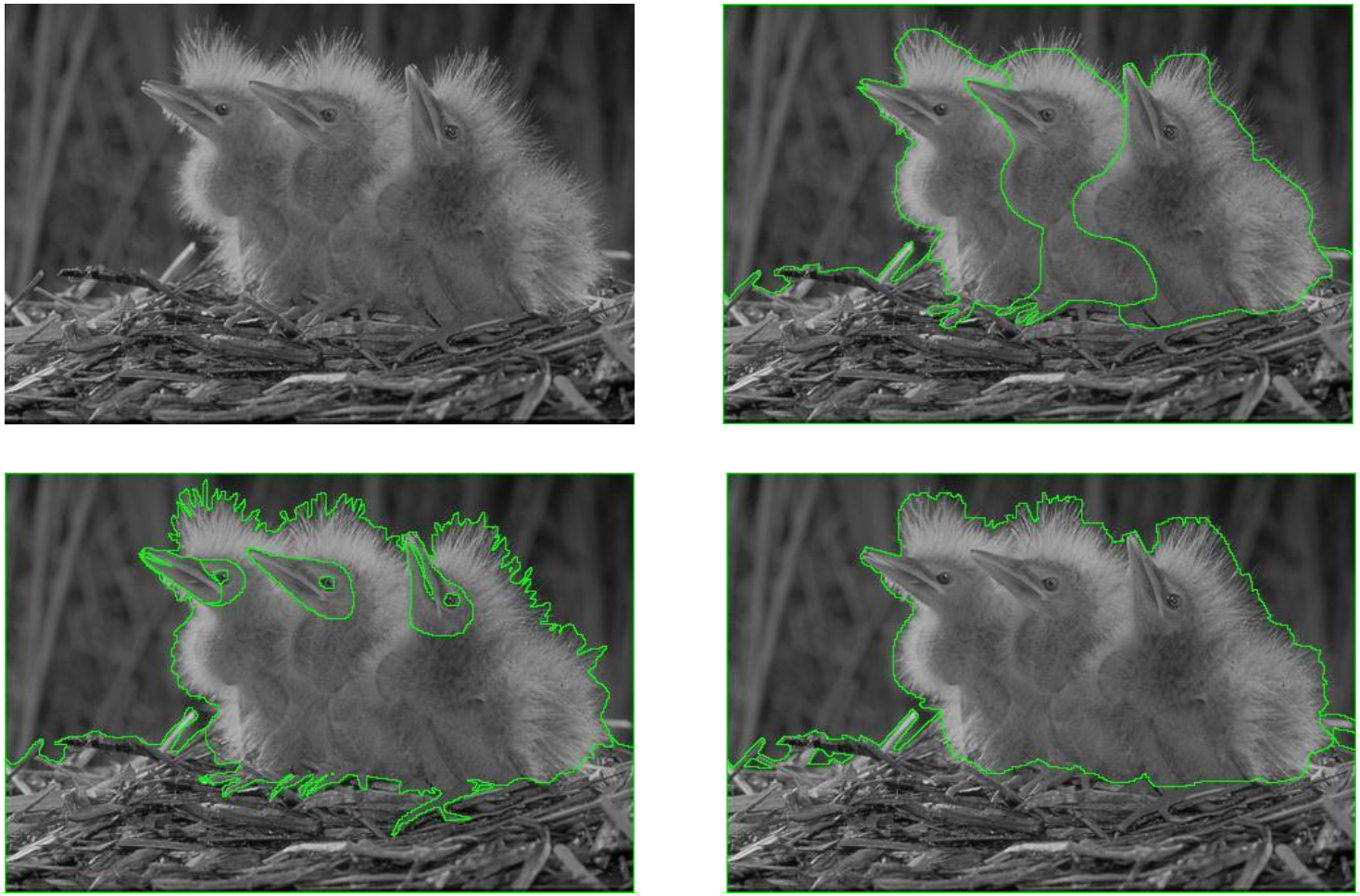
Segmentation

Segmentation is conceptually ill defined



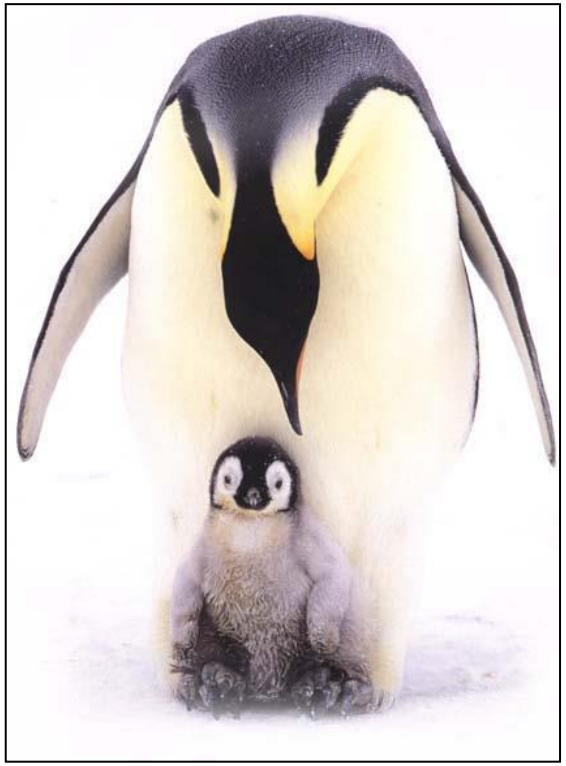
Segmentation

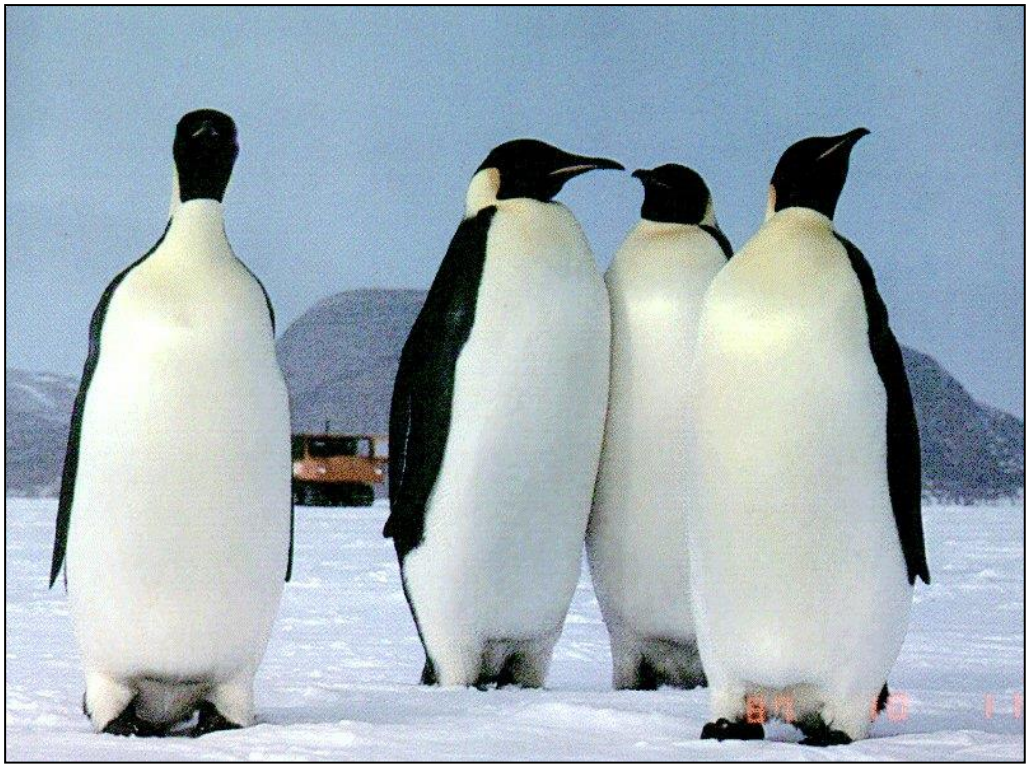
Segmentation is difficult!!



Segmentation

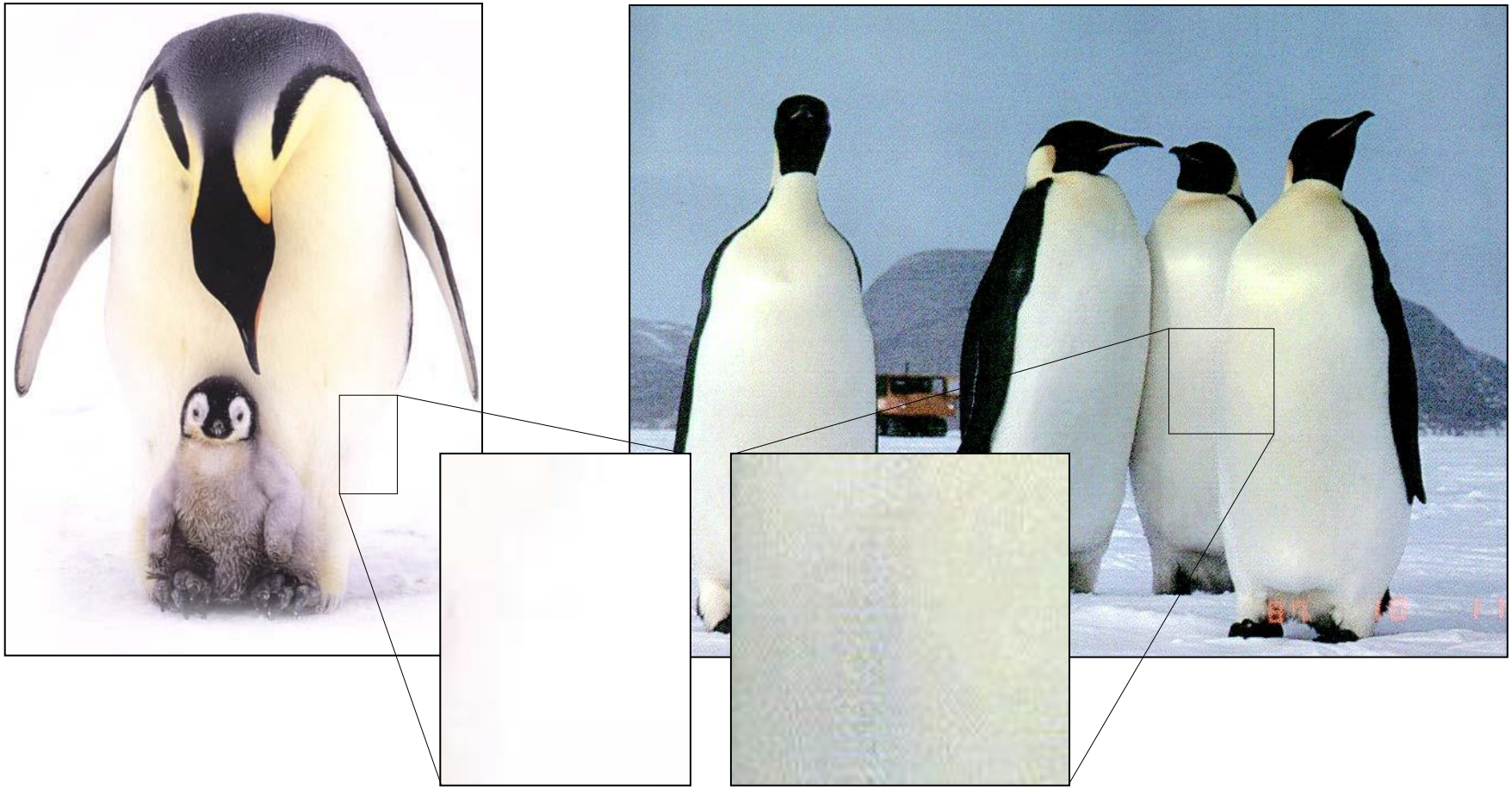
Segmentation is difficult!!





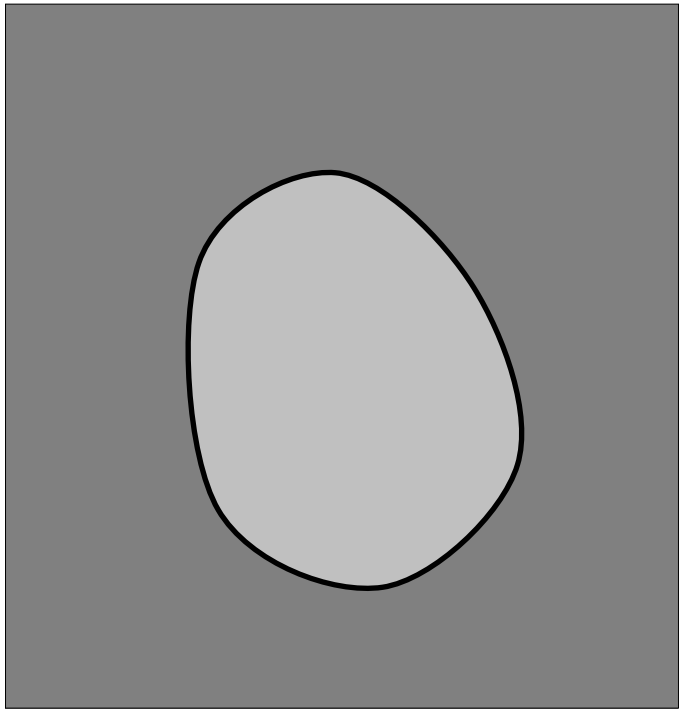
Segmentation

Segmentation is difficult!!



Segmentation

Segmentation is dual to boundary/edge detection



Segmentation (Regions)

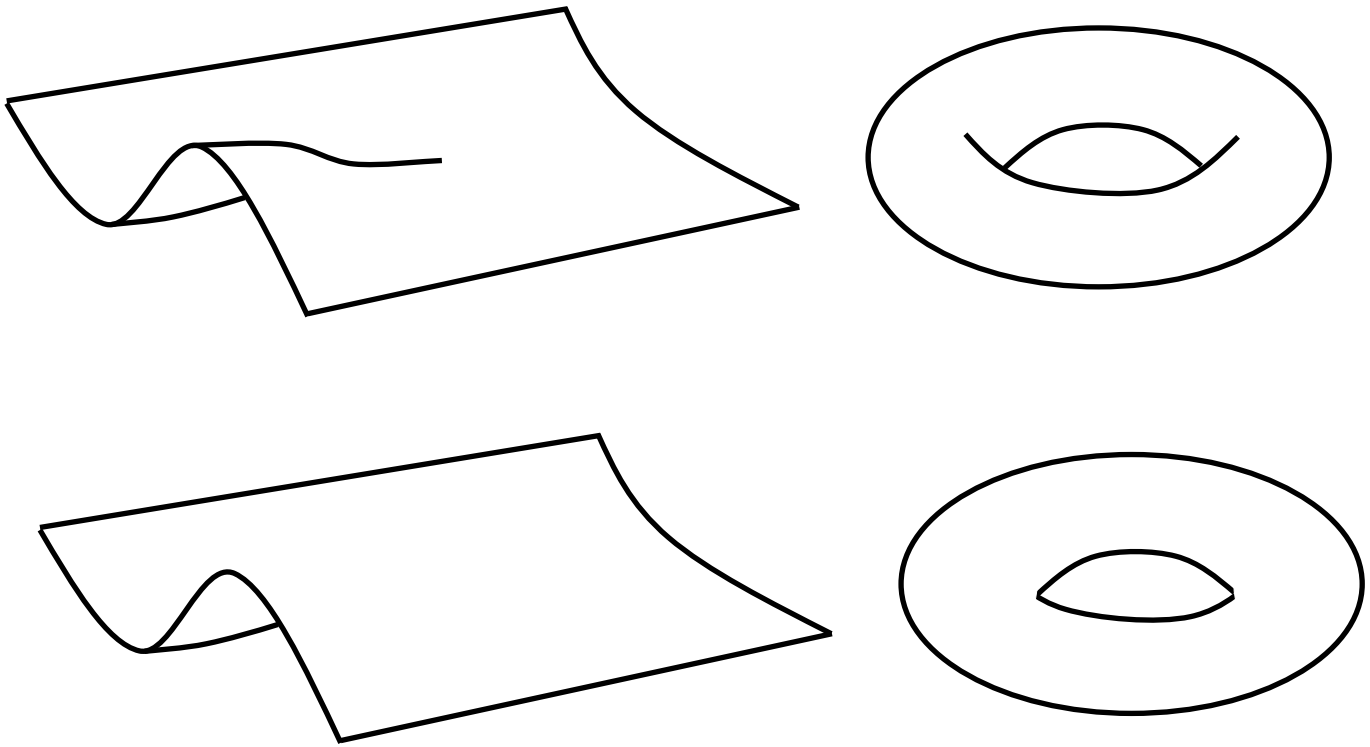
Make explicit intra-region coherence

Edges (Boundaries)

Make explicit inter-region differences

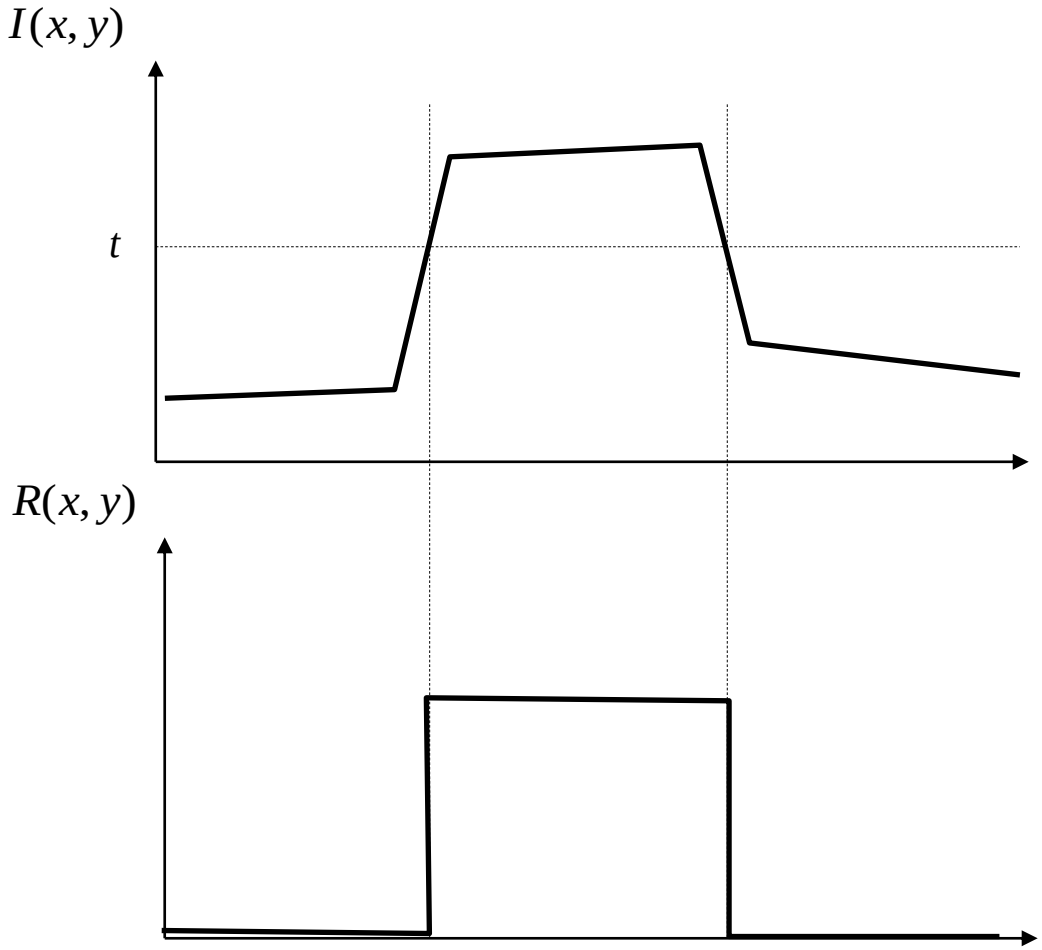
Segmentation

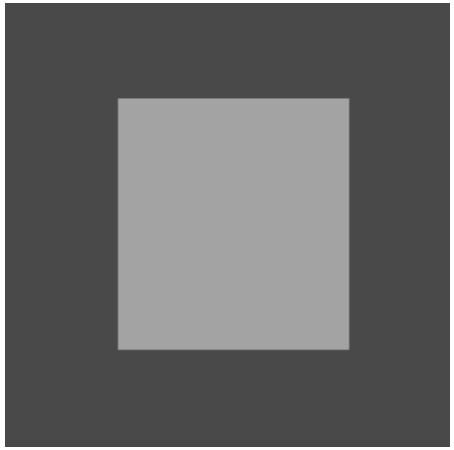
Is segmentation dual to boundary/edge detection???



Segmentation Approaches

Segmentation via thresholding

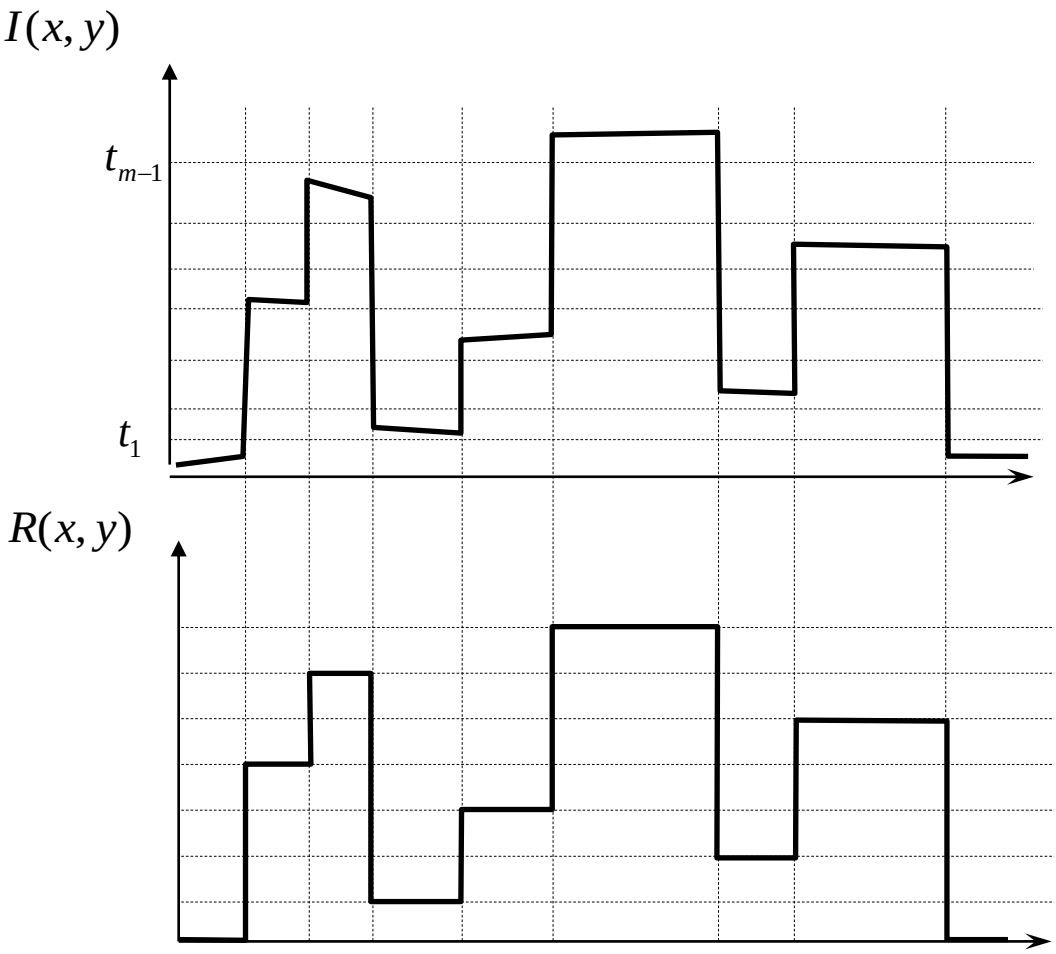


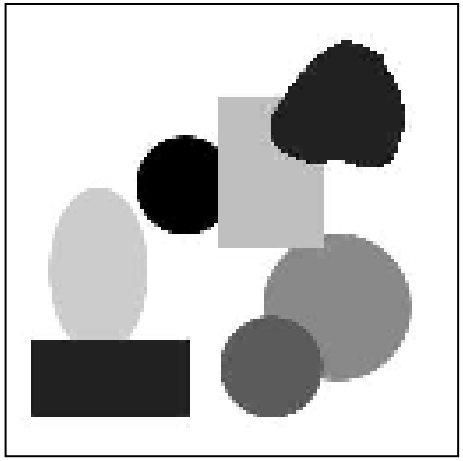


$$R(x, y) = \begin{cases} 0 & I(x, y) < t \\ 1 & t \le I(x, y) \end{cases}$$

Segmentation Approaches

Segmentation via thresholding

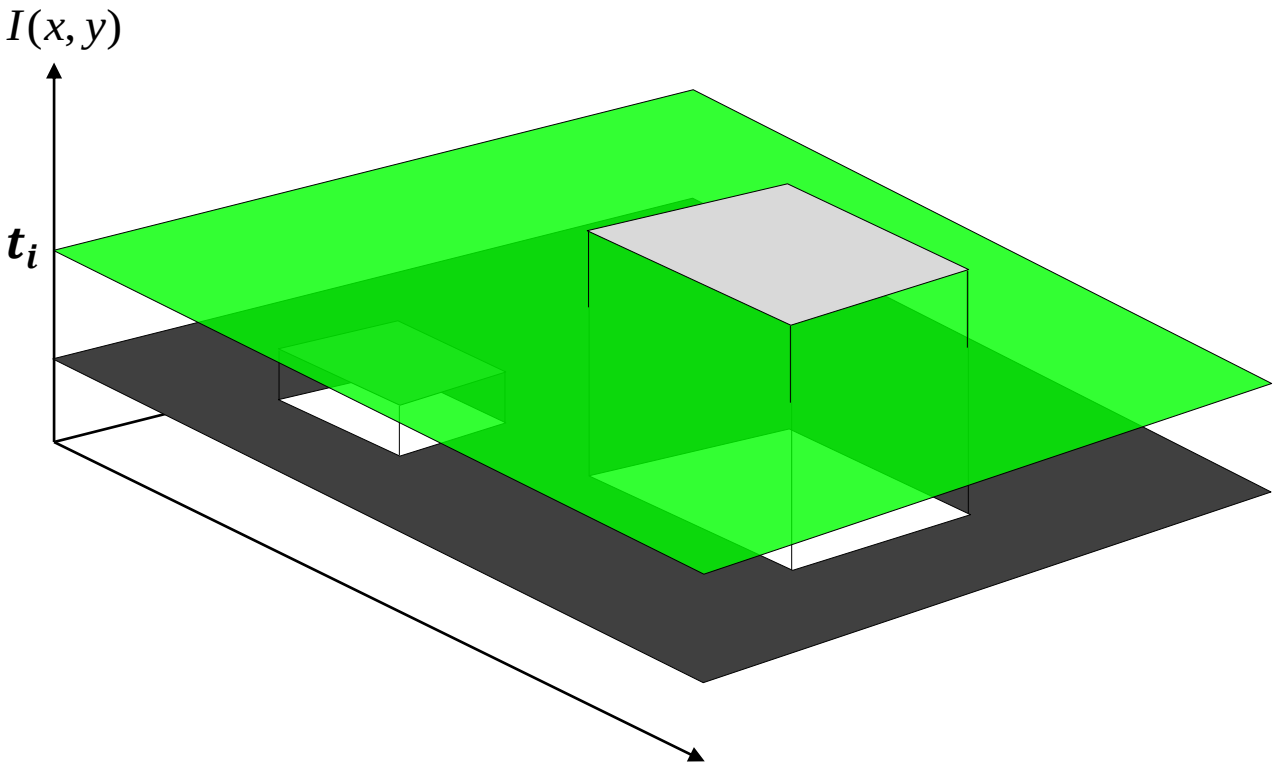




$$R(x,y) = \begin{cases} 1 & I_{\min} \le I(x,y) < t_1 \\ 2 & t_1 \le I(x,y) < t_2 \\ 3 & t_2 \le I(x,y) < t_3 \\ \vdots & \vdots \\ m & t_{m-1} \le I(x,y) < I_{\max} \end{cases}$$

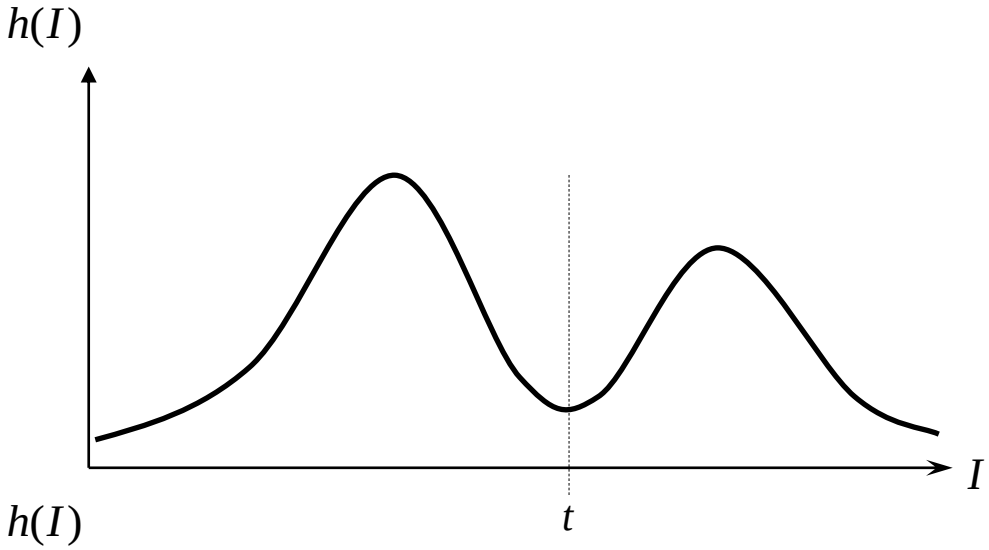
Segmentation Approaches

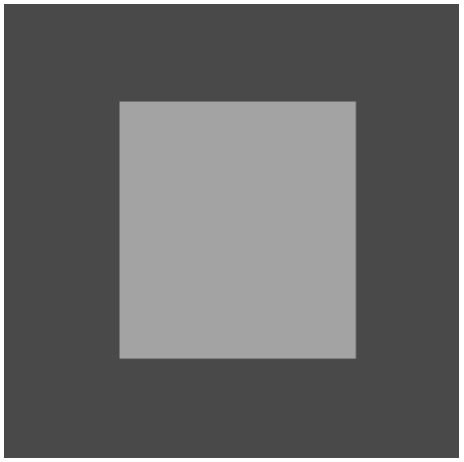
Global thresholding

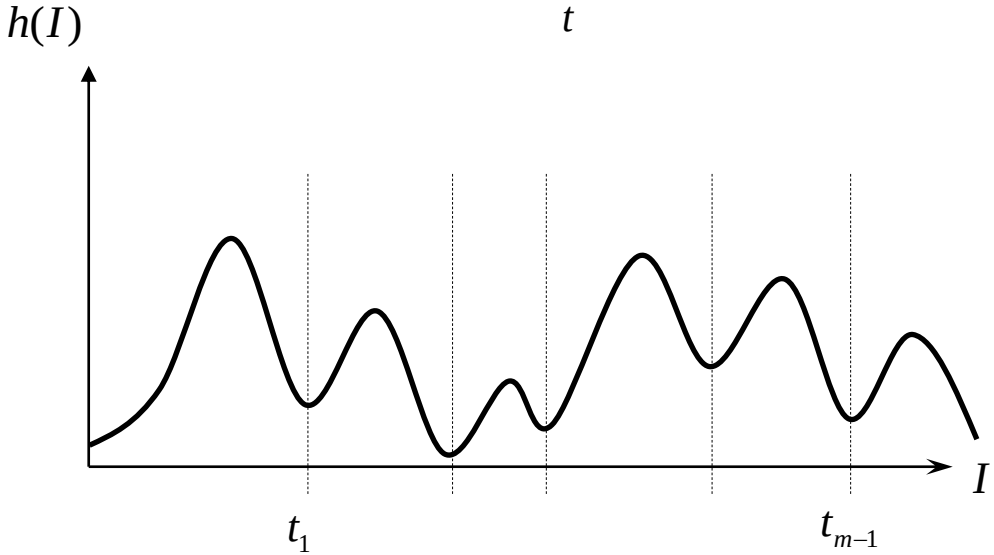


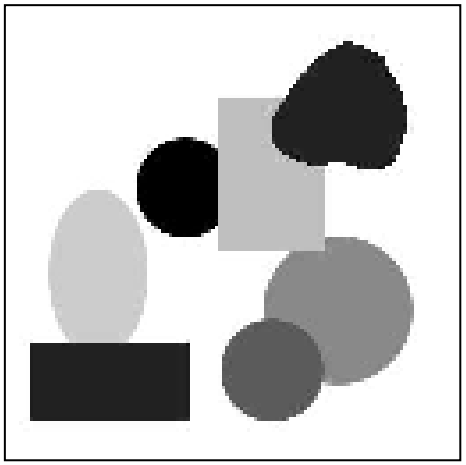
Segmentation Approaches

Automatic global threshold selection



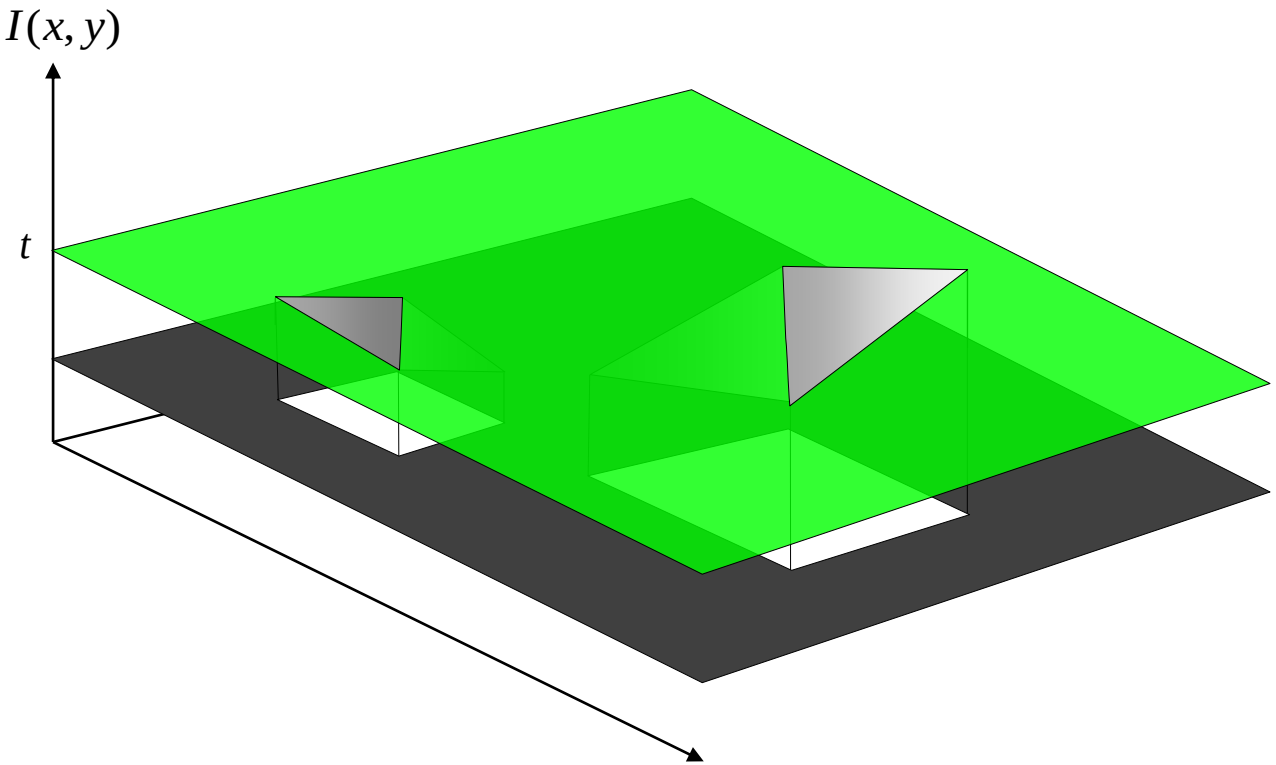




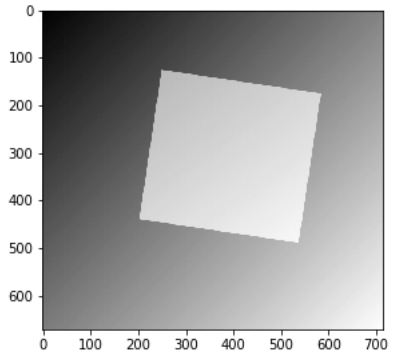


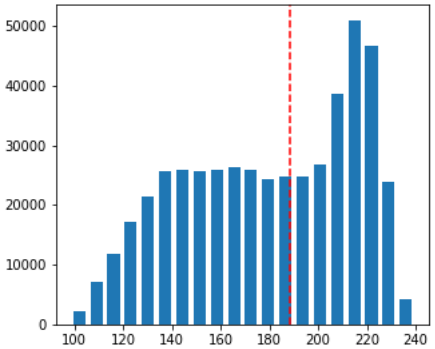
Segmentation Approaches

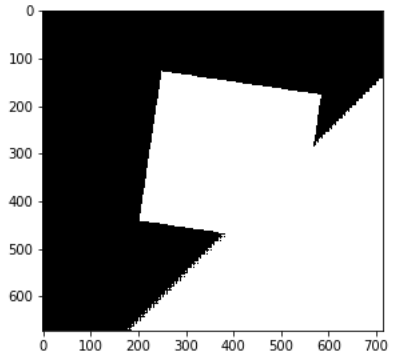
Global thresholding? Watch out



Segmentation Approaches







In real images

- **Noise**
- **gradual changes in illumination**

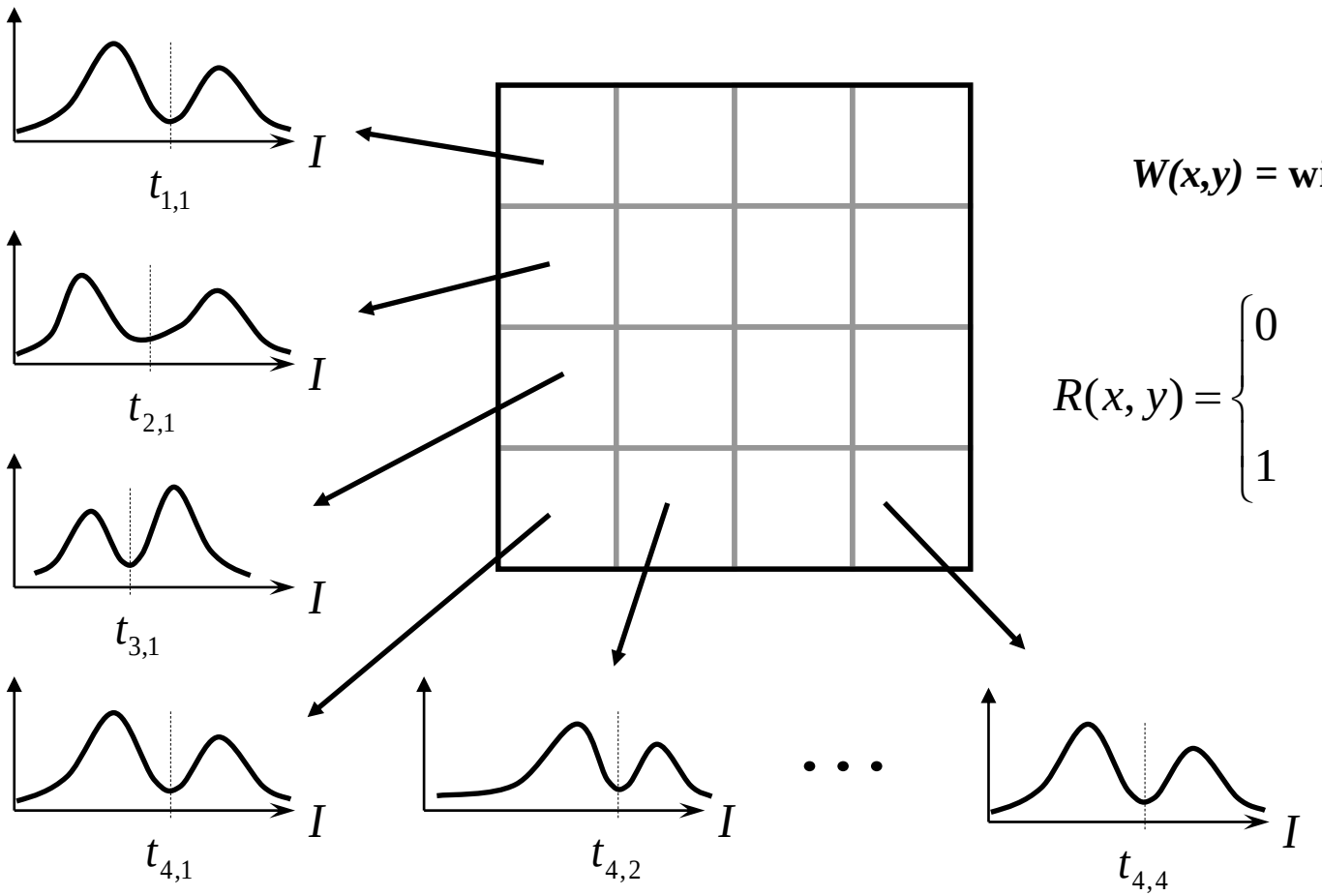
Global thresholding is likely to fail

Possible improvement:

Local adaptive thresholding

Segmentation Approaches

Local adaptive thresholding

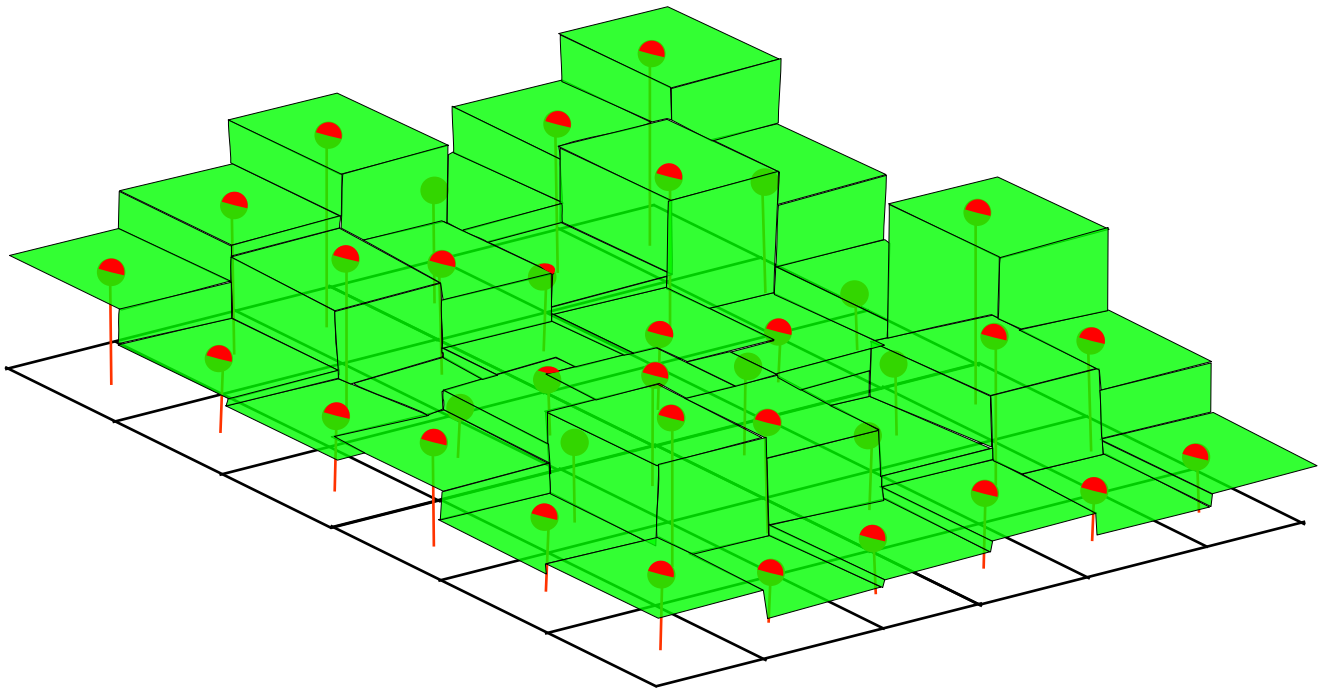


***W(x,y)* = window of pixel (x,y)**

$$R(x,y) = \begin{cases} 0 & I(x,y) < t_{W(x,y)} \\ 1 & t_{W(x,y)} \le I(x,y) \end{cases}$$

Segmentation Approaches

Adaptive thresholding



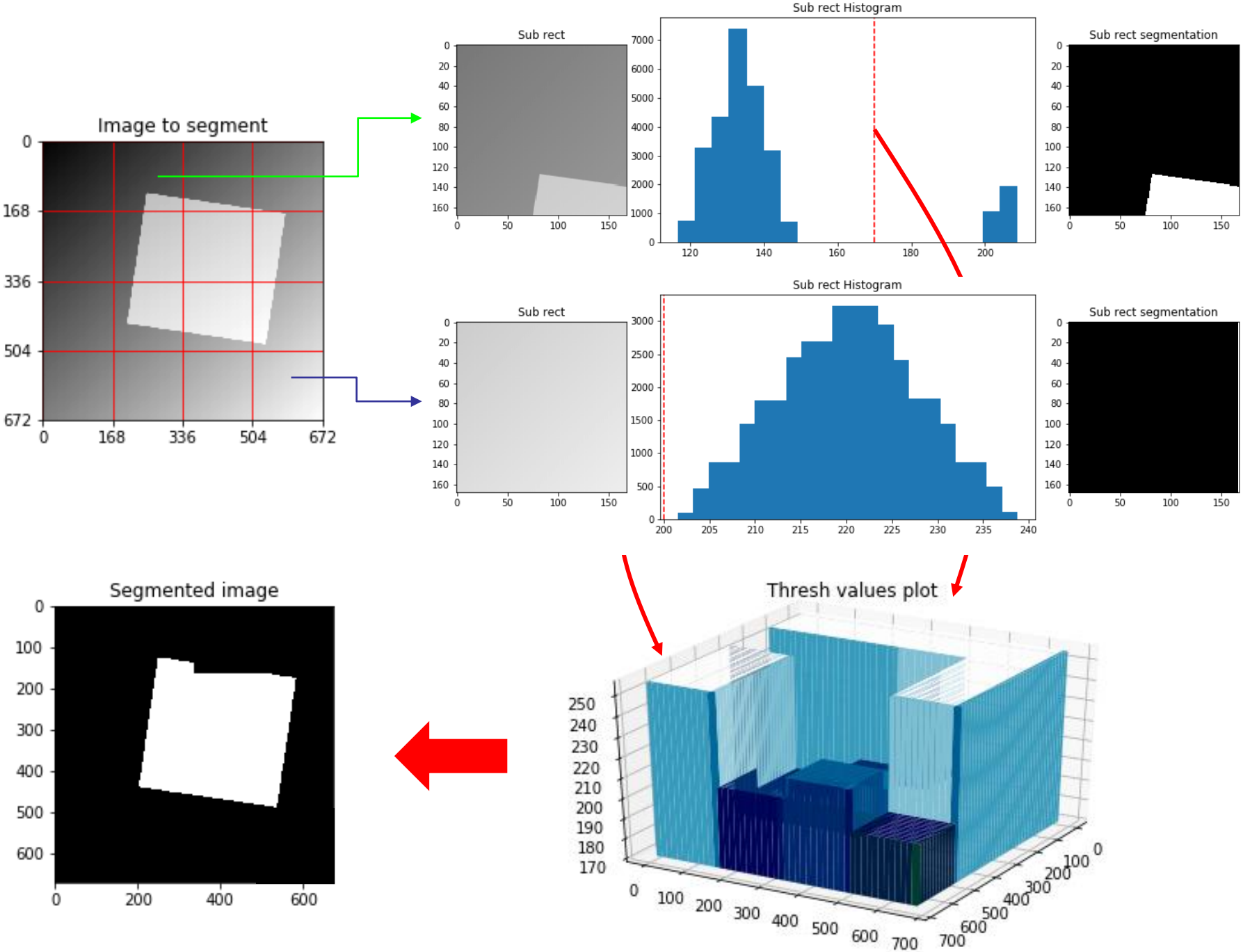
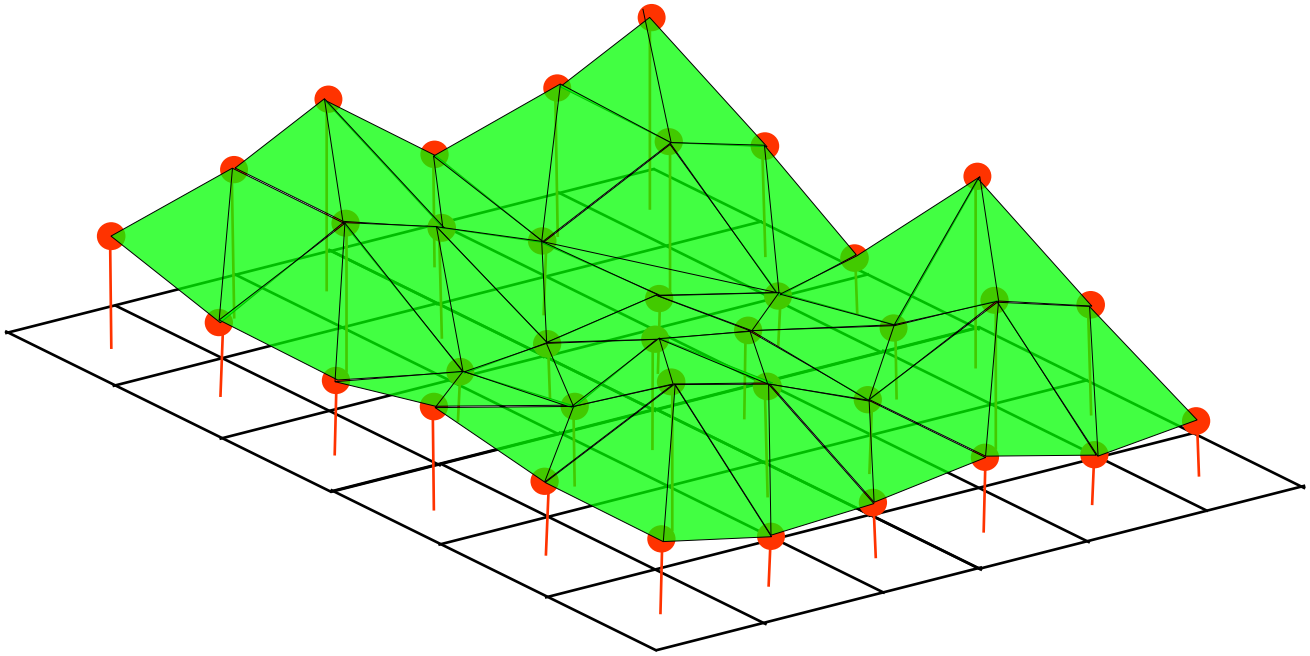
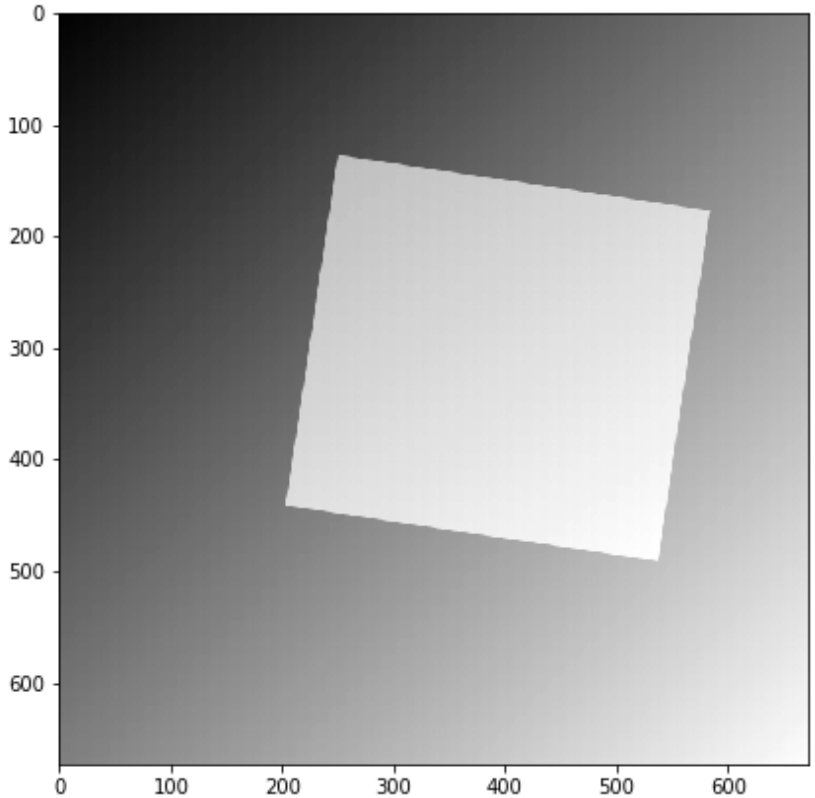


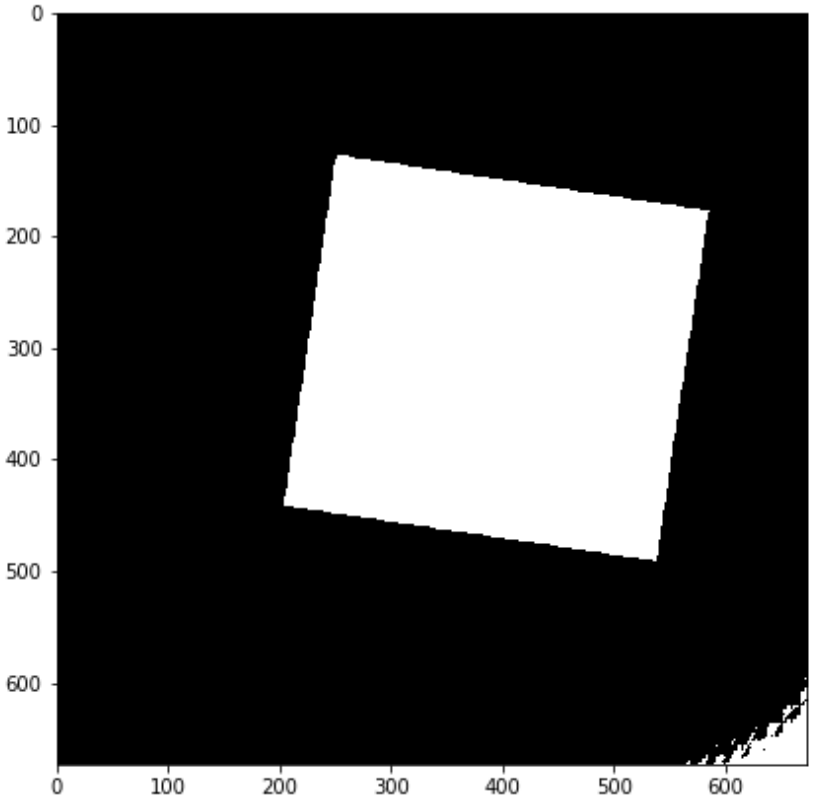
Image to segment
Sub rect
Sub rect Histogram
Sub rect segmentation
Sub rect
Sub rect Histogram
Sub rect segmentation
Segmented image
Thresh values plot

Segmentation Approaches

Adaptive thresholding

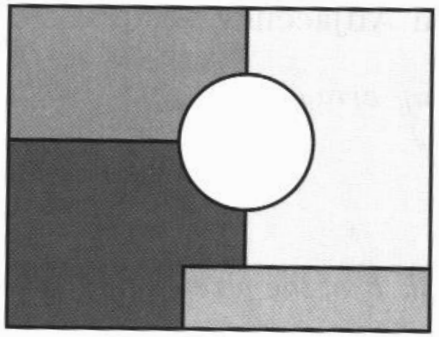


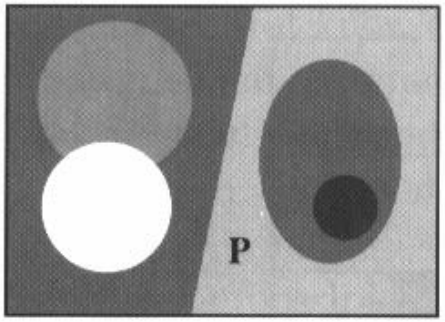

0
100
200
300
400
500
600
0
100
200
300
400
500
600

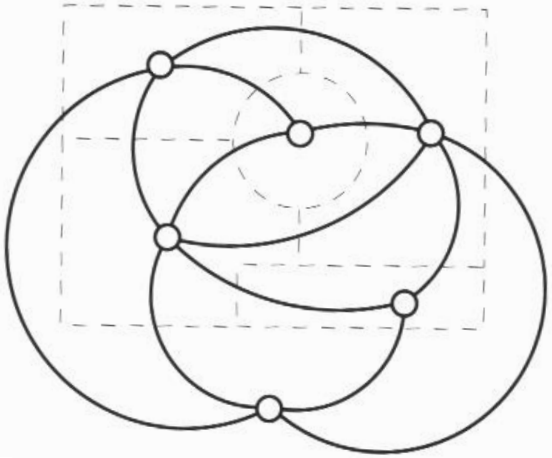

0
100
200
300
400
500
600
0
100
200
300
400
500
600

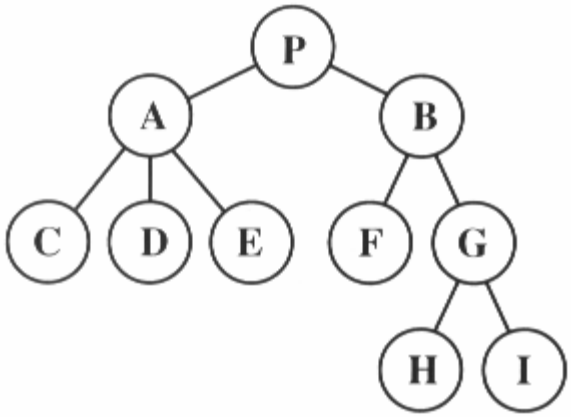
Segmentation Approaches

Representing segmentations







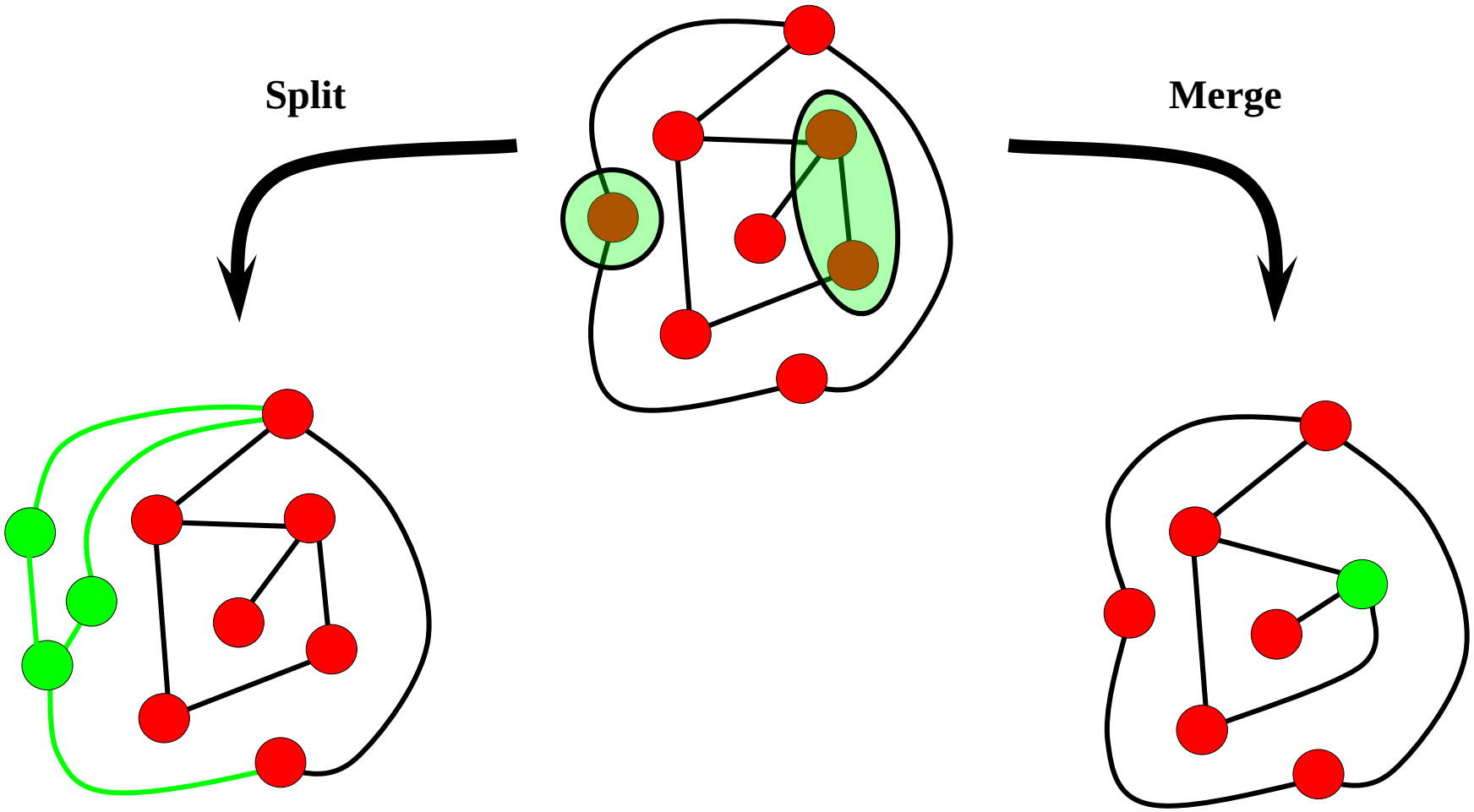


Region Adjacency Graphs (RAGs)

Region (Picture) Trees

Segmentation Approaches

Split and Merge



Segmentation Approaches

Region merging

Region merging

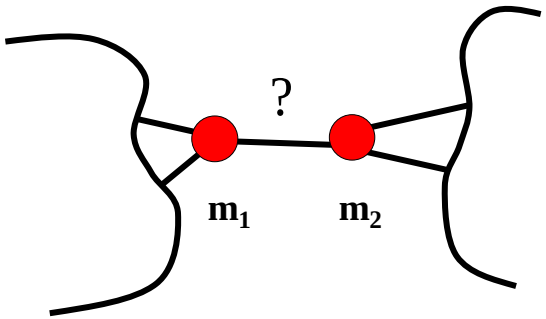
1. Form initial segmentation
2. Compute RAG
3. Repeat
   - Pick an edge e that connects two regions R_i and R_j in the RAG
   - If $\boldsymbol{H}(\boldsymbol{R_i} \cup \boldsymbol{R_j}) = \boldsymbol{true}$ then merge the two regions and update the RAG

   Until no more regions can be merged

Segmentation Approaches

Merging statistically similar regions

Assumption: region have constant feature value corrupted by statistically independent, additive, normally distributed noise.



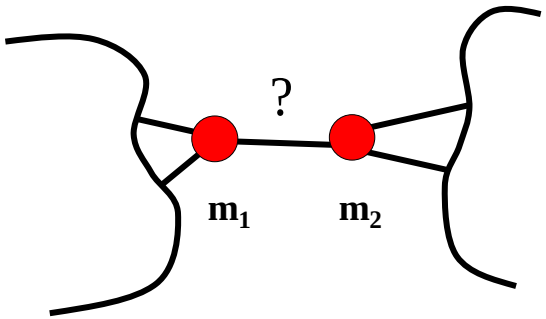
Hypothesis $\mathbf{H_0}$: Regions should be merged. Their feature values are all drawn from the same single normal distribution with parameters (μ_0, σ_0)

Hypothesis $\mathbf{H_1}$: Regions should **not** be merged. Their feature values are drawn from two different normal distributions with parameters (μ_1, σ_1) and (μ_2, σ_2)

Which hypothesis should be selected ?

Segmentation Approaches

Merging statistically similar regions



Probability of any given value

$$P(v_i) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(v_i-\mu)^2}{2\sigma^2}}$$

Distribution mean

$$\hat{\mu} = \frac{1}{n}\sum_i v_i$$

Distribution variance

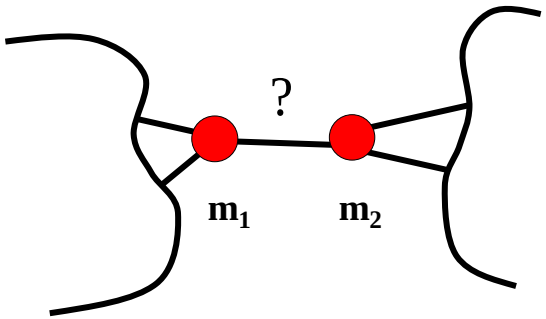
$$\hat{\sigma}^2 = \frac{1}{n}\sum_i (v_i - \hat{\mu})^2$$

$$P(v_1, v_2, \dots, v_{m_1+m_2}|H_0) = \prod_{i=1}^{m_1+m_2} P(v_i|H_0) = \prod_{i=1}^{m_1+m_2} \frac{1}{\sqrt{2\pi}\sigma_0} e^{-\frac{(v_i-\mu_0)^2}{2\sigma_0^2}} = \frac{1}{\left(\sqrt{2\pi}\sigma_0\right)^{m_1+m_2}} e^{-\frac{\sum_{i=1}^{m_1+m_2}(v_i-\mu_0)^2}{2\sigma_0^2}} = \frac{1}{\left(\sqrt{2\pi}\sigma_0\right)^{m_1+m_2}} e^{-\frac{m_1+m_2}{2}}$$

$$P(v_1, v_2, \dots, v_{m_1+m_2}|H_1) = \prod_{i=1}^{m_1+m_2} P(v_i|H_1) = \prod_{i=1}^{m_1} \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(v_i-\mu_1)^2}{2\sigma_1^2}} \cdot \prod_{i=m_1+1}^{m_2} \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(v_i-\mu_2)^2}{2\sigma_2^2}} = \cdots = \frac{1}{\left(\sqrt{2\pi}\sigma_1\right)^{m_1}\left(\sqrt{2\pi}\sigma_2\right)^{m_2}} e^{-\frac{m_1+m_2}{2}}$$

Segmentation Approaches

Merging statistically similar regions



Probability of any given value

$$P(v_i) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(v_i-\mu)^2}{2\sigma^2}}$$

Distribution mean

$$\hat{\mu} = \frac{1}{n}\sum_i v_i$$

Distribution variance

$$\hat{\sigma}^2 = \frac{1}{n}\sum_i (v_i - \hat{\mu})^2$$

Liklihood ratio

$$L = \frac{P\left(v_1, v_2, \dots, v_{m_1+m_2} \middle| H_1\right)}{P\left(v_1, v_2, \dots, v_{m_1+m_2} \middle| H_0\right)} = \frac{\sigma_0^{m_1+m_2}}{\sigma_1^{m_1} \cdot \sigma_2^{m_2}}$$

Segmentation Approaches

Region splitting

Region splitting

1. Form initial segmentation
2. Compute RAG
3. Repeat
   - Pick a node R_i in the RAG
   - If $H(R_i) = false$ then split the region and update the RAG

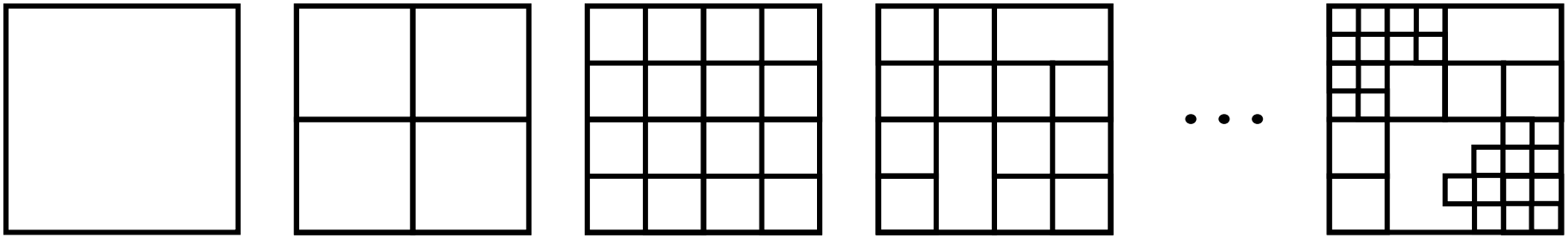
   Until no more splits can be done

Segmentation Approaches

Split and Merge

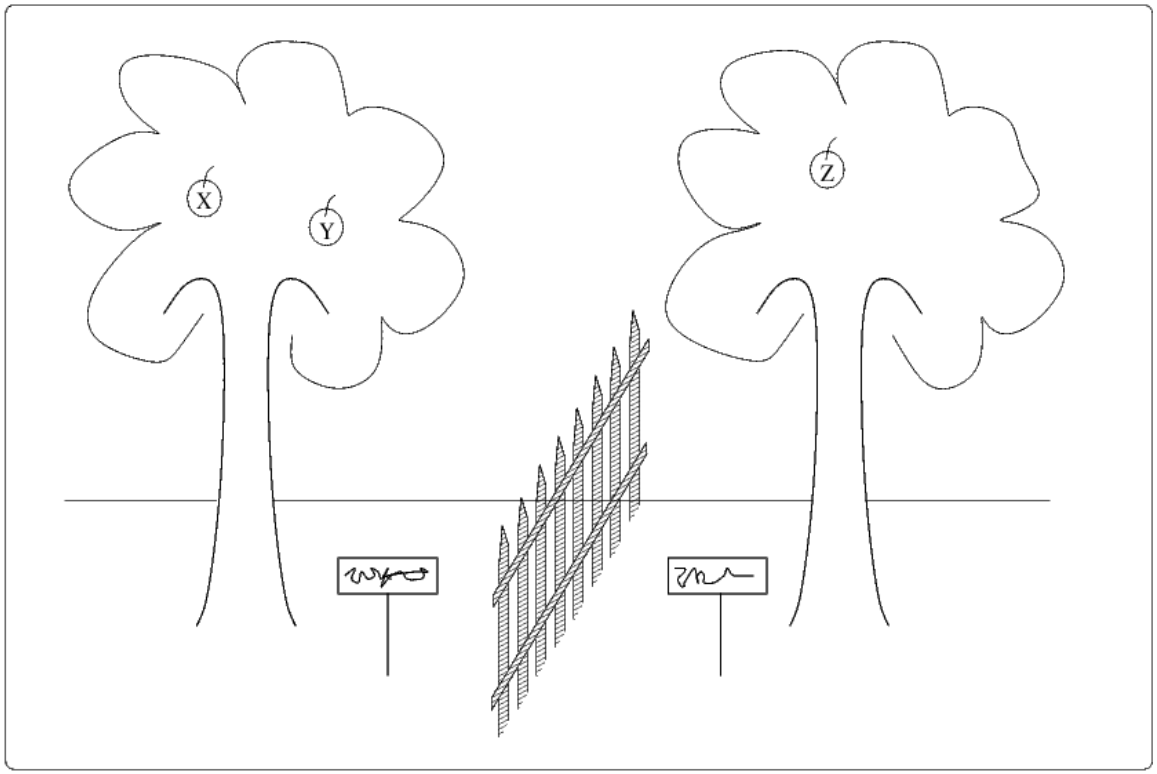
1. Form initial segmentation
2. Compute RAG
3. Repeat
   - Pick a node R_i in the RAG and examine it for splitting. Update RAG if split is exercised.
   - Pick an edge e that connects two regions *Ri* and *Rj* in the RAG and examine it for merging. Update RAG if merge is exercised.

   Until no more splits can be done



Segmentation Approaches

Segmentation via relaxation



Contextual constraints

$$X = Y$$

$$X \neq Z$$

$$Y \neq Z$$

$$\downarrow$$

$$X = apple$$

$$Y = apple$$

$$Z = orange$$

Measurements

$P_X(\text{Apple}) = 100\%$ $P_Y(\text{Apple}) = 50\%$ $P_Z(\text{Apple}) = 50\%$

$P_X(\text{Orange}) = 0\%$ $P_Y(\text{Orange}) = 50\%$ $P_Z(\text{Orange}) = 50\%$

Segmentation Approaches

Segmentation via relaxation

$B = \{b_1, \ldots, b_n\}$ set of **objects** to be labeled

$\Lambda = \{1, 2, \ldots, m\}$ set of possible **labels**

$p_i^0(\lambda)$ the measured **confidence** that b_i should be labeled λ.

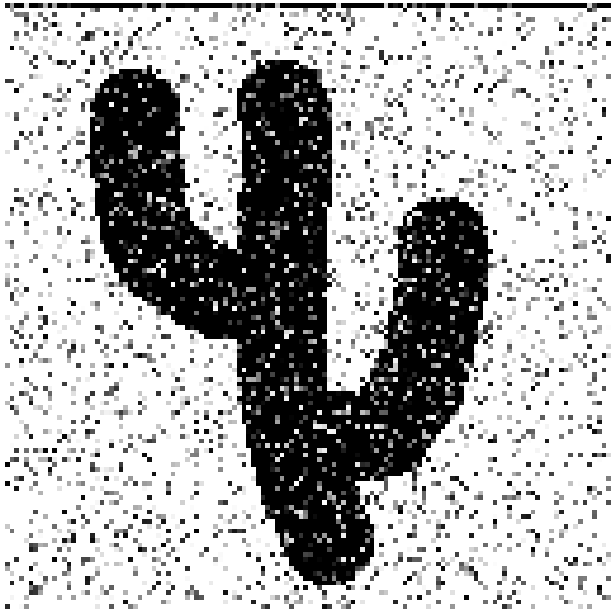
$p_i^0(\lambda) \geq 0$

$\sum_{\lambda=1}^{m} p_i^0(\lambda) = 1 \quad \forall i$

$r_{ij}(\lambda, \mu) \quad \Leftrightarrow$ the strength of **compatibility** between the hypotheses "b_i has label λ" and "b_j has label μ"

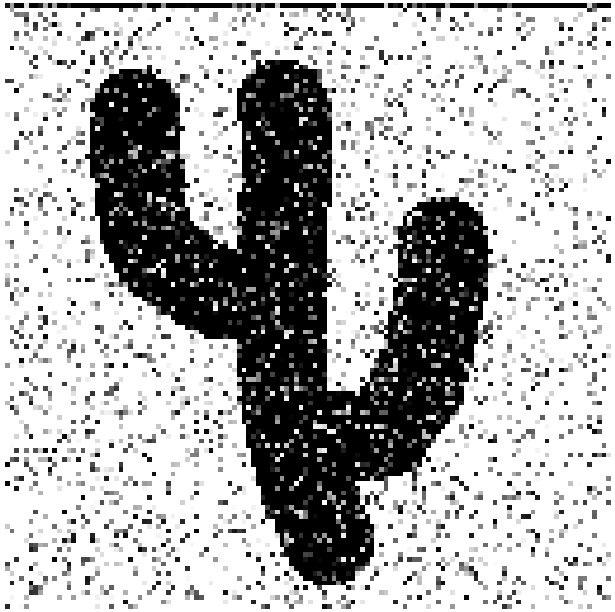
Segmentation Approaches

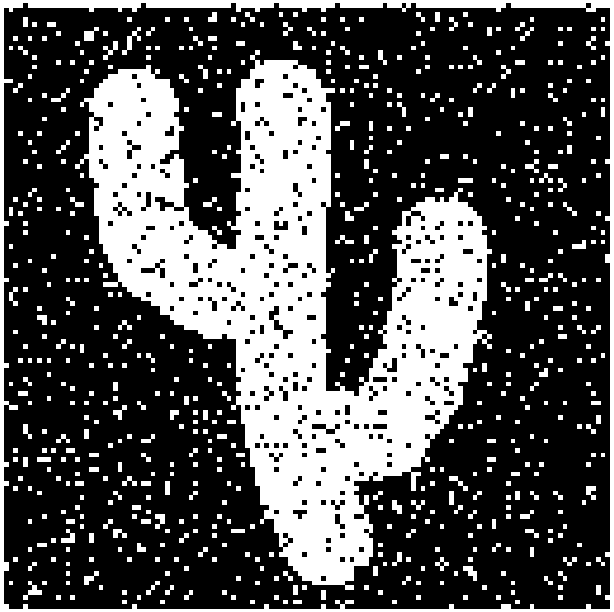
Segmentation via relaxation



Segmentation Approaches

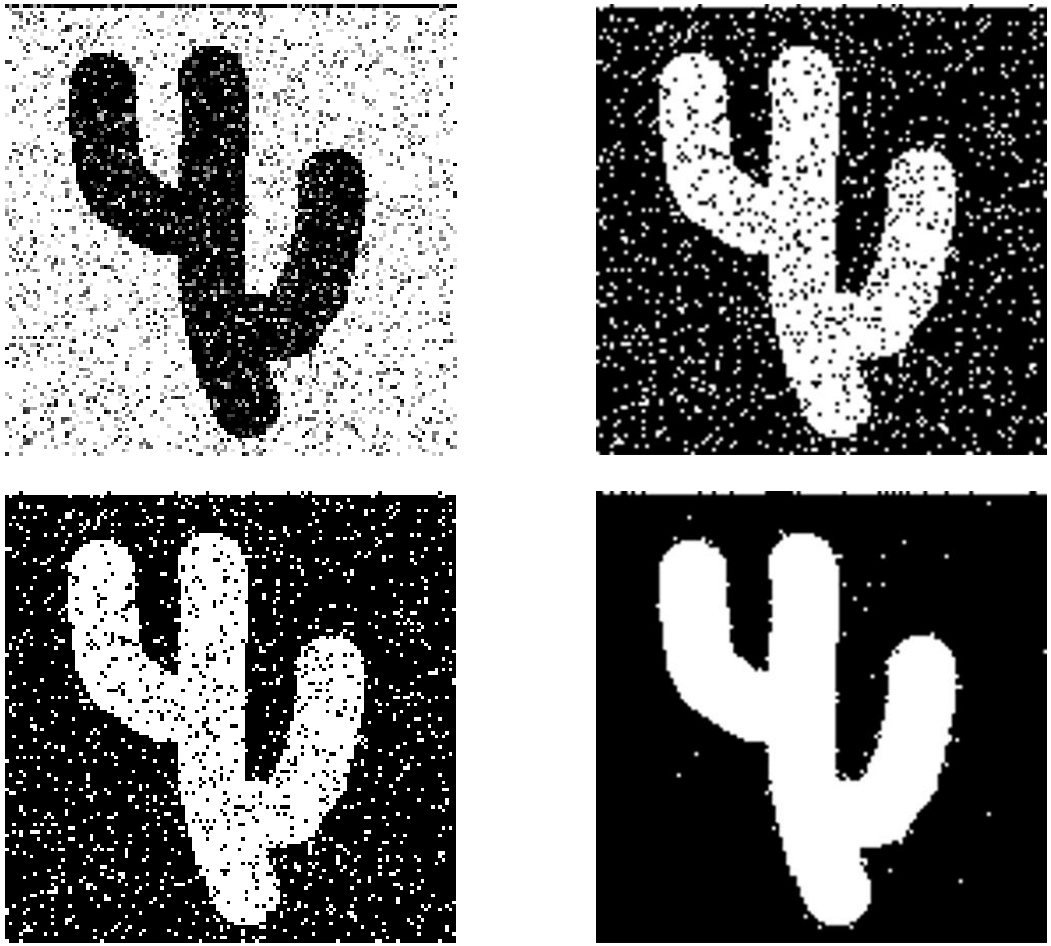
Segmentation via relaxation





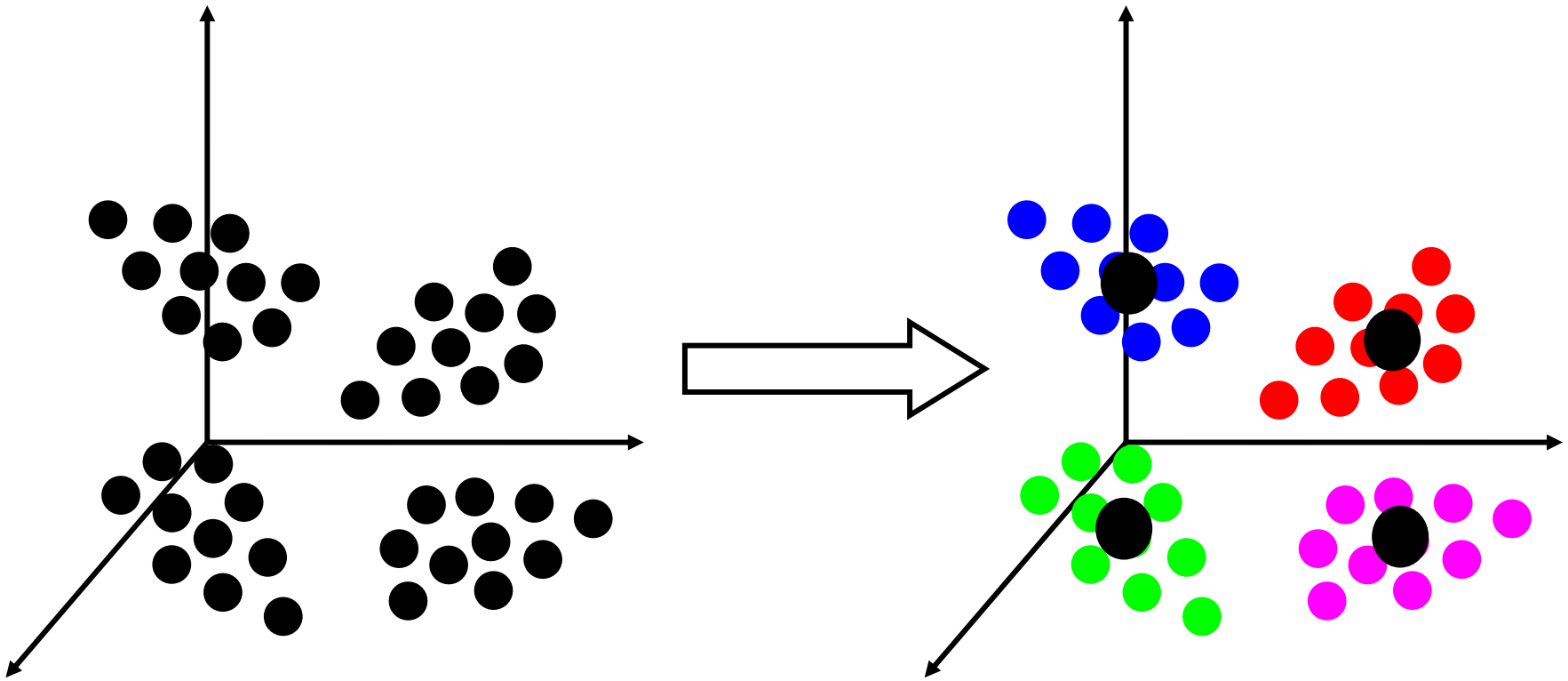
Segmentation Approaches

Segmentation via relaxation



Segmentation Approaches

Segmentation via clustering



Segmentation Approaches

Segmentation via clustering

Problem formulation:

Given a set of data points x_i, find K clusters C_j, with representatives m_j, such that the total fit measure of data points to clusters is minimized

$$D = \sum_{k=1}^{K} \sum_{x_i \in C_k} d(x_i, m_i) \rightarrow \min$$

Least square error measure: $d(x_i, m_i) = \left\| x_i - m_i \right\|^2$

Image segmentation as clustering:

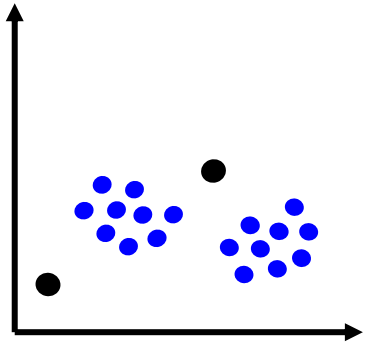
- $\boldsymbol{x_i}$: feature vectors associated with pixels
- $\boldsymbol{C_j}$: segments
- $\boldsymbol{m_j}$: representative (mean) feature vector for sement j

Segmentation Approaches

Iterative K-means clustering

1. Chose randomly the set of K cluster centers $m_1, m_2, \ldots, m_K$
2. Repeat
   - Allocate each data point to the cluster whose center is nearest
   - Update all m_i to the center point of their cluster

   Until cluster centers are unchanged



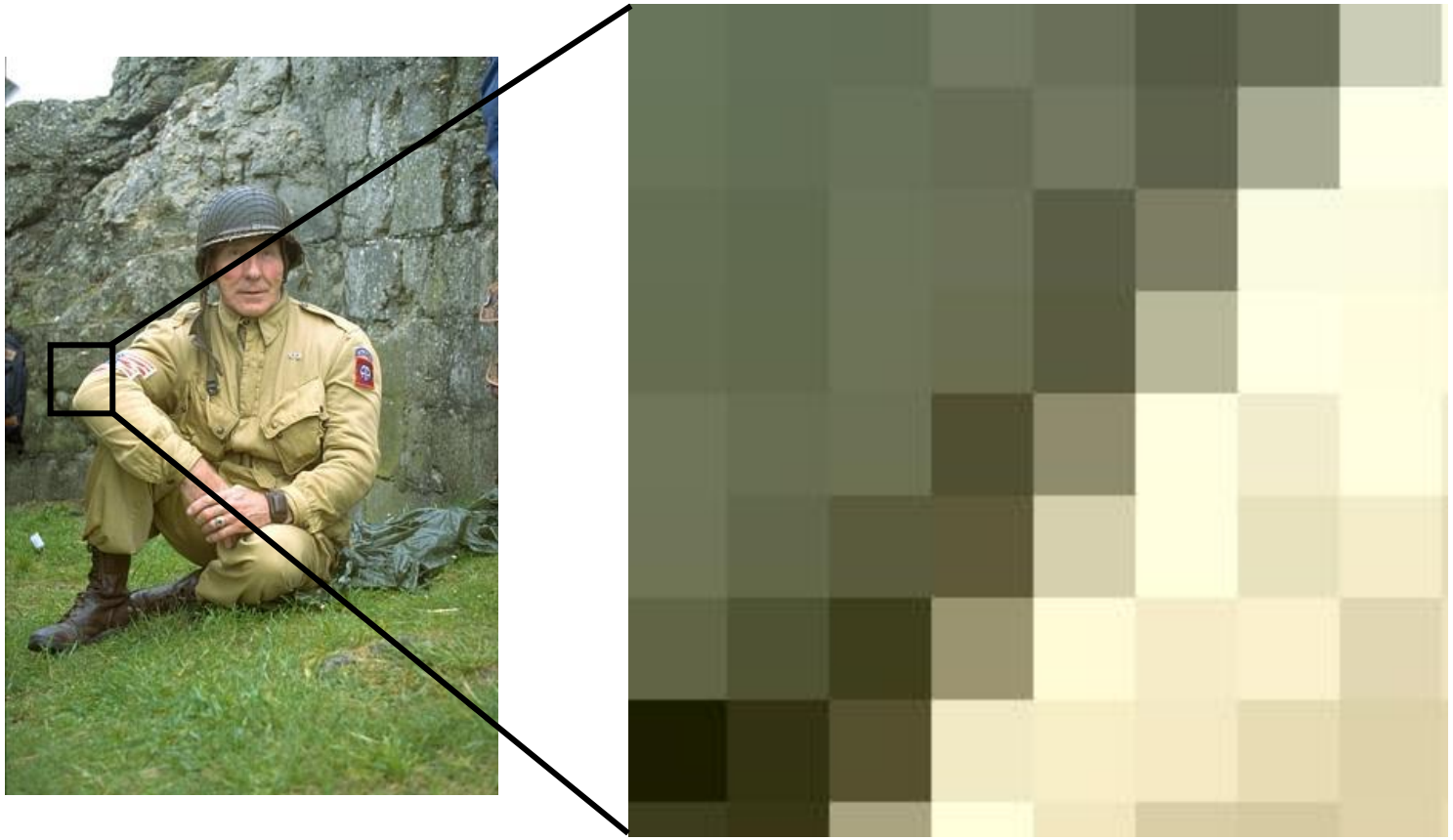
Segmentation Approaches

Iterative K-means clustering

6-clusters

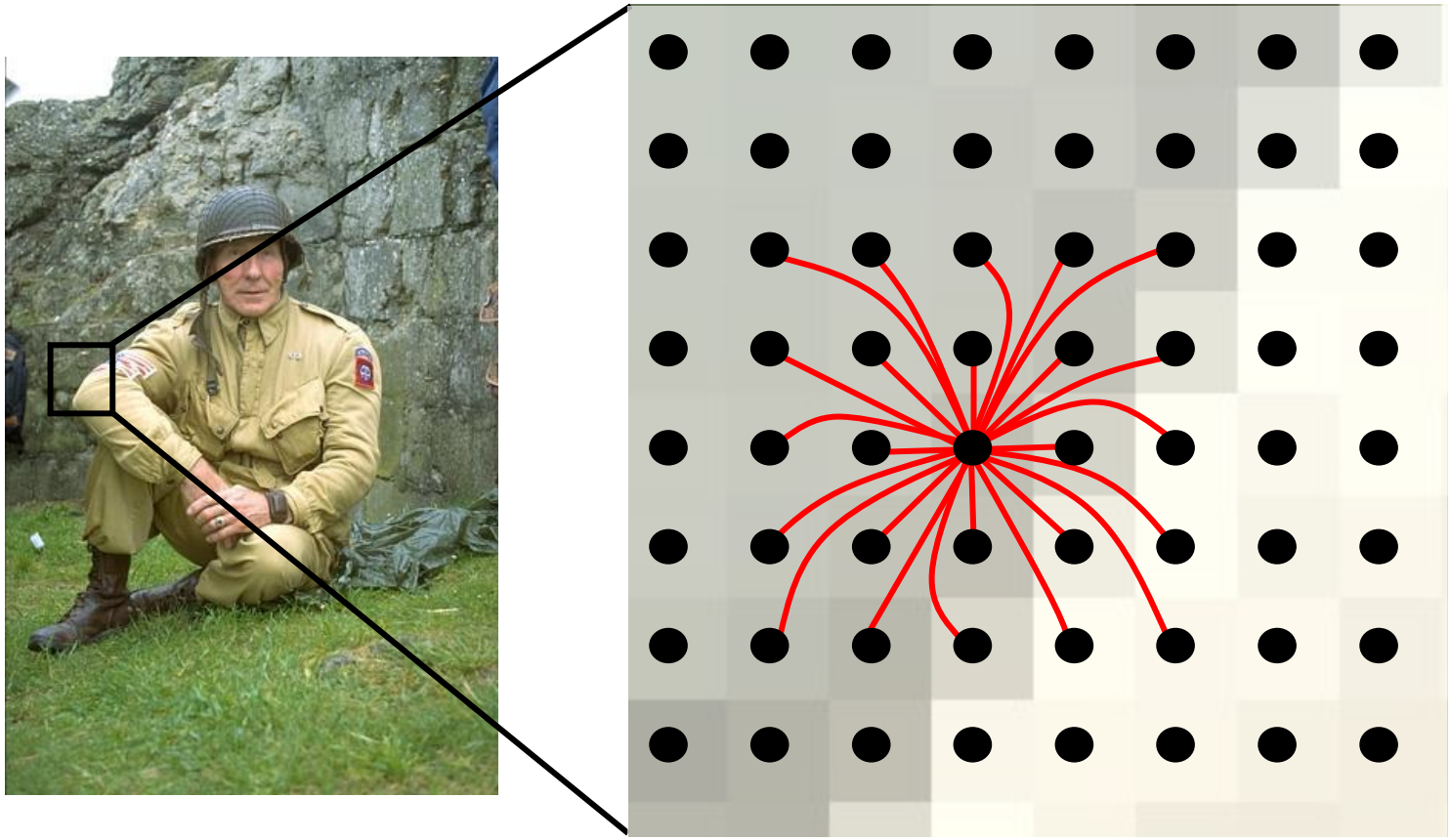
Segmentation Approaches

Graph theoretic approach to segmentation



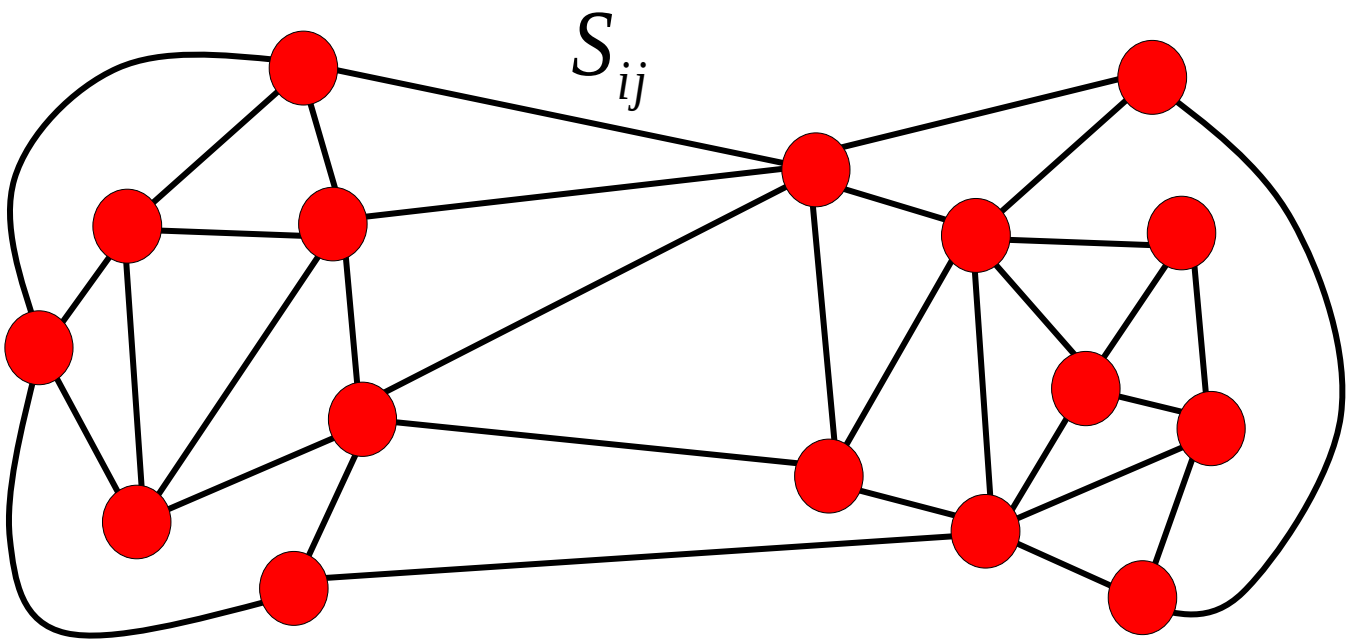
Segmentation Approaches

Graph theoretic approach to segmentation



Segmentation Approaches

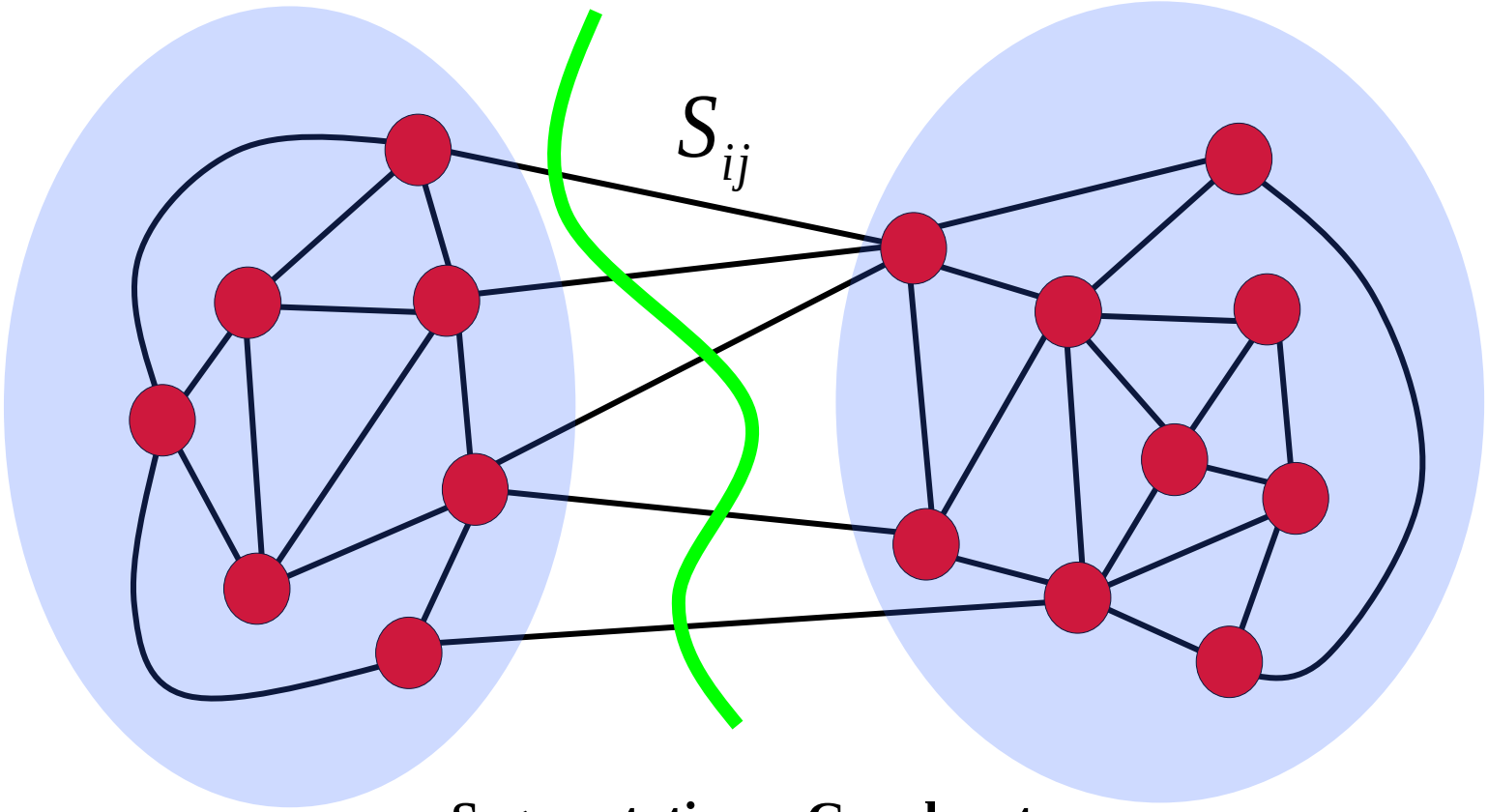
Graph theoretic approach to segmentation



S_{ij} = similarity (weight/affinity) based on intensity, color, texture, etc…

Segmentation Approaches

Graph theoretic approach to segmentation



Segmentation = Graph cuts

Segmentation Approaches

Graph theoretic approach to segmentation

Given:

a graph representation *(V,E)* of the image and a pairwise similarity measure

Compute:

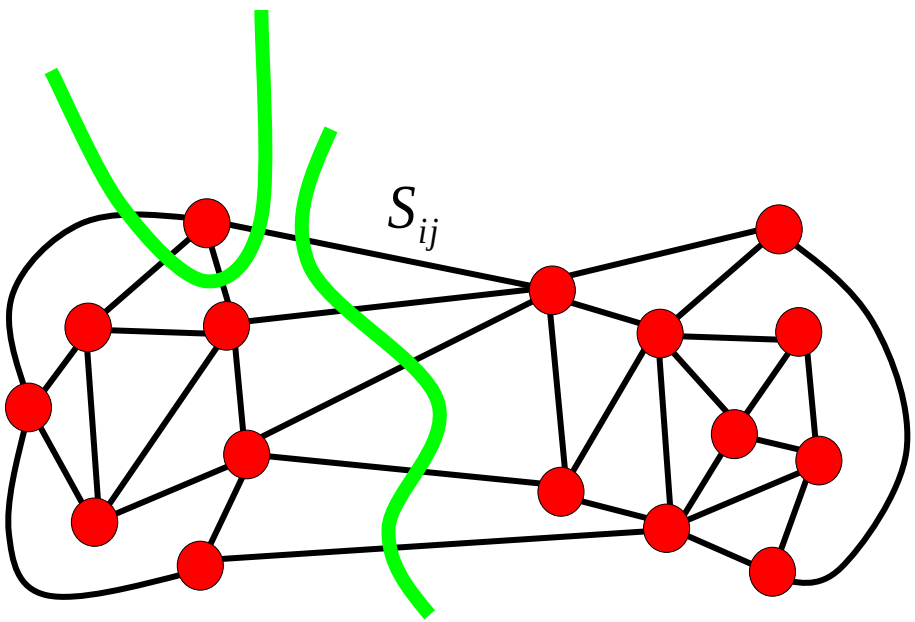
A partition of the graph into disjoint sets $V_1, V_2, \ldots V_m$ such that the total similarity is maximized within each V_i and is minimized between any two V_i anf V_j

Recursive formulation:

A partition of the graph into disjoint sets *A* and *B* and apply recursively to *A* and *B*.

Segmentation Approaches

Graph theoretic approach to segmentation



$$A \cup B = V$$

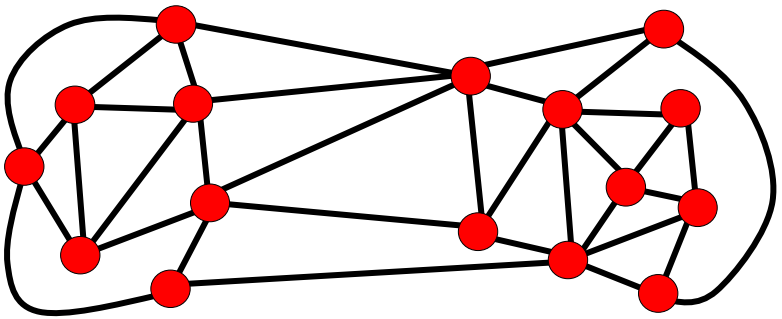
$$A \cap B = \varnothing$$

$$cut(A,B) = \sum_{e_1 \in A; e_2 \in B} S_{e_1,e_2}$$

Segmentation $\overset{?}{=}$ Minimum cuts

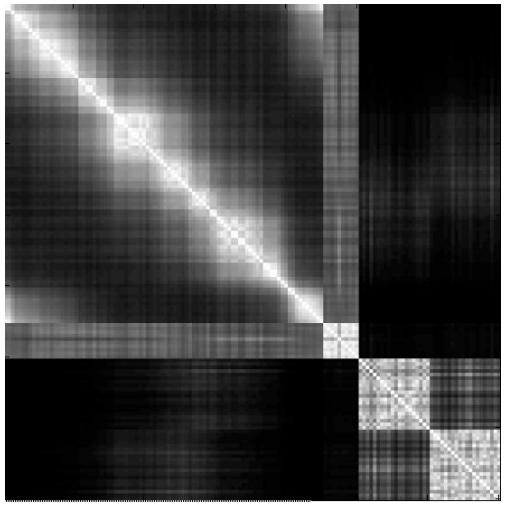
Segmentation Approaches

Graph theoretic approach to segmentation



$$cut(A,B) = \sum_{e_1 \in A; e_2 \in B} S_{e_1,e_2}$$

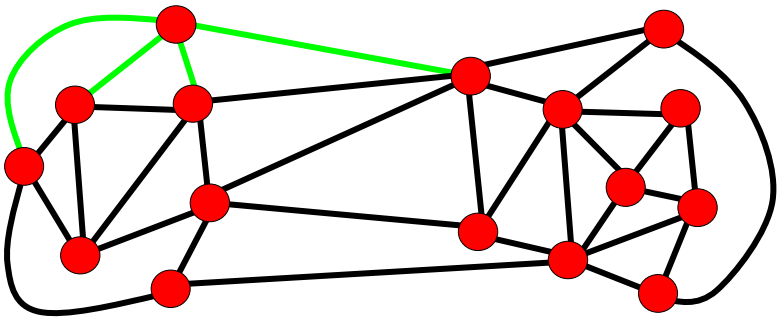
$$W = [S_{ij}]$$



Similarity Matrix

Segmentation Approaches

Graph theoretic approach to segmentation

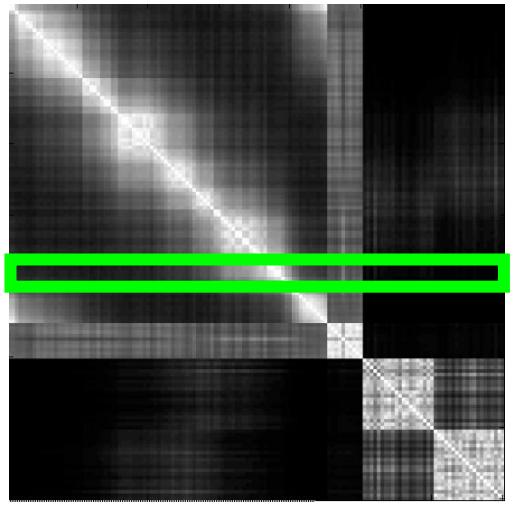


$$cut(A,B) = \sum_{e_1 \in A; e_2 \in B} S_{e_1,e_2}$$

$$W = [S_{ij}]$$

$$d_i = \sum_j S_{ij}$$

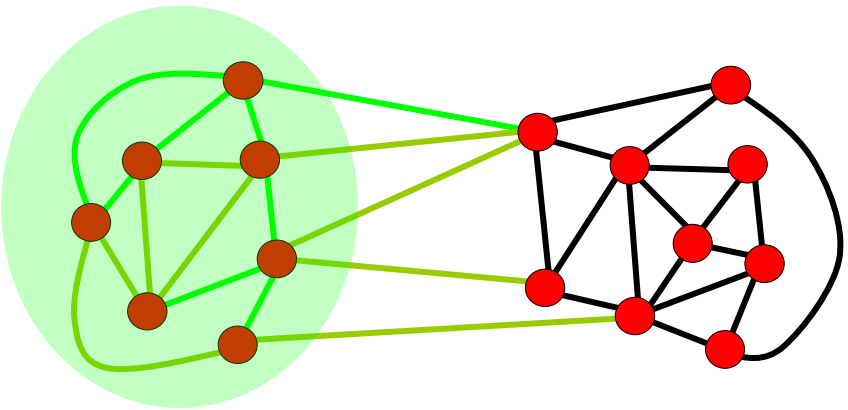
Degree of a node

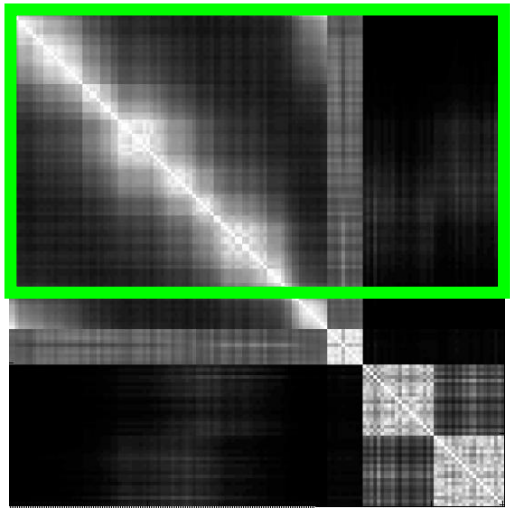


Similarity Matrix

Segmentation Approaches

Graph theoretic approach to segmentation





Similarity Matrix

$$cut(A,B) = \sum_{e_1 \in A; e_2 \in B} S_{e_1,e_2}$$

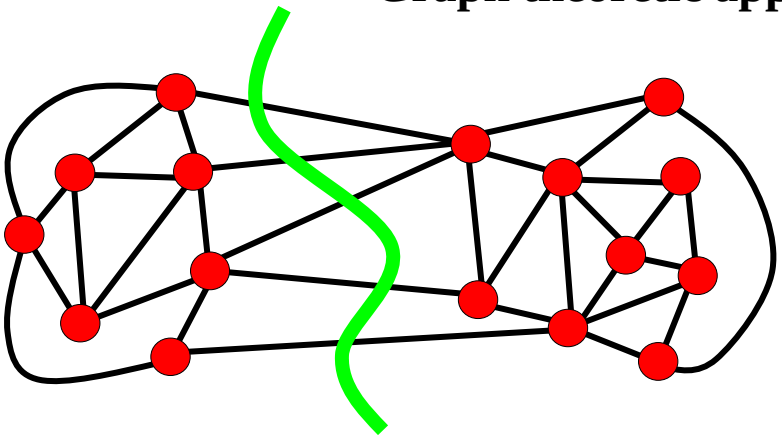
$$W = [S_{ij}]$$

$$d_i = \sum_j S_{ij}$$ **Degree of a node**

$$vol(A) = \sum_{i \in A} d_i$$ **Volume of a set**

Segmentation Approaches

Graph theoretic approach to segmentation



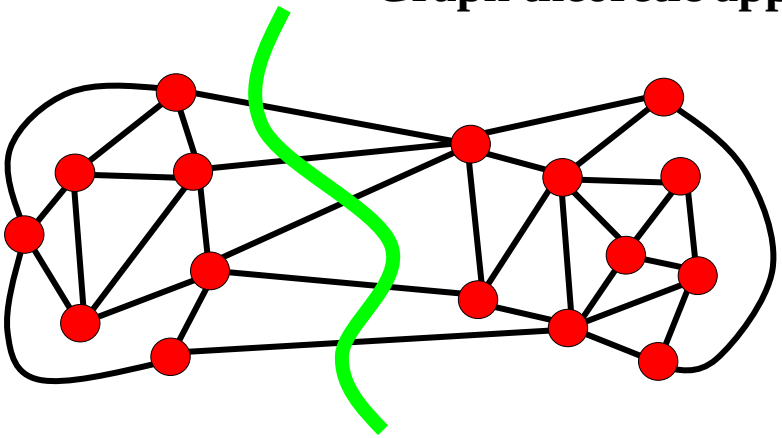
Normalized cuts

$$Ncut(A,B) = \frac{cut(A,B)}{vol(A)} + \frac{cut(B,A)}{vol(B)}$$

NP-Hard!! ☹

Segmentation Approaches

Graph theoretic approach to segmentation



$$Ncut(A,B) = \frac{cut(A,B)}{vol(A)} + \frac{cut(B,A)}{vol(B)}$$

Degree Matrix

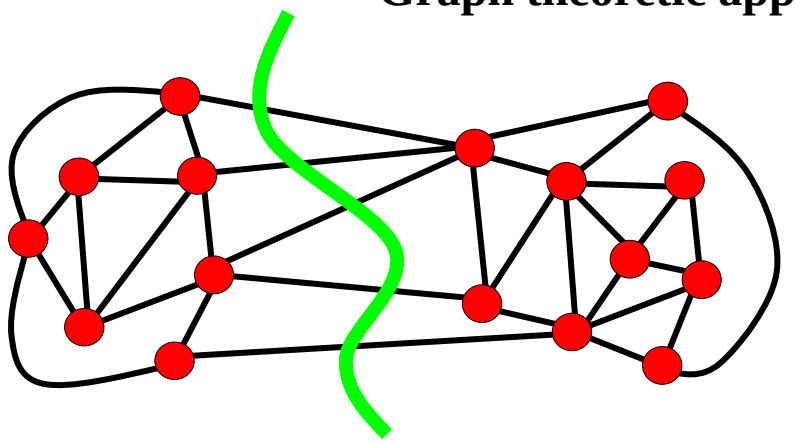
$$D = \begin{bmatrix} d_1 & 0 & 0 & 0 \\ 0 & d_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & d_n \end{bmatrix}$$

Laplacian Matrix $L = D - W$

Segmentation vector $\vec{x} = (x_i)$ $\quad x_i = \begin{cases} 1 & i \in A \\ -1 & i \notin A \end{cases}$

Segmentation Approaches

Graph theoretic approach to segmentation



$$Ncut(A,B) = \frac{cut(A,B)}{vol(A)} + \frac{cut(B,A)}{vol(B)}$$

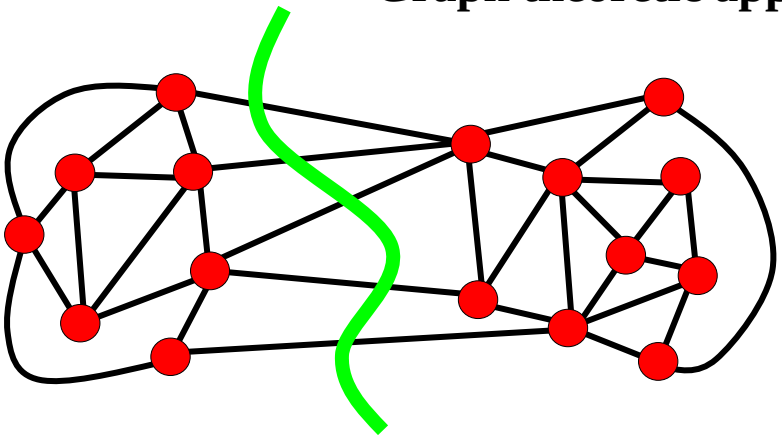
$$Ncut(A,B) \to \min \quad \Leftrightarrow \quad \frac{y^T \overbrace{(D-W)}^{L} y}{y^T D y} \to \min$$

$$y = (1+x) - b(1-x) \qquad b = \frac{k}{1-k} \qquad k = \frac{\sum_{x_i > 0} d_i}{\sum_i d_i}$$

$$y_i = \begin{cases} 2 & i \in A \\ -2b & i \notin A \end{cases}$$

Segmentation Approaches

Graph theoretic approach to segmentation



$$Ncut(A,B) = \frac{cut(A,B)}{vol(A)} + \frac{cut(B,A)}{vol(B)}$$

$$Ncut(A,B) \rightarrow \min \quad \Leftrightarrow \quad \frac{y^T \overbrace{(D-W)}^{L} y}{y^T D y} \rightarrow \min$$

Relax

(allow any *y*)

$$\overbrace{(D-W)}^{L} y = \lambda D y$$

$$z = D^{\frac{1}{2}} y \quad \Leftrightarrow \quad D^{-\frac{1}{2}} L D^{-\frac{1}{2}} z = \lambda z \quad \Leftrightarrow \quad Mz = \lambda z$$

Segmentation Approaches

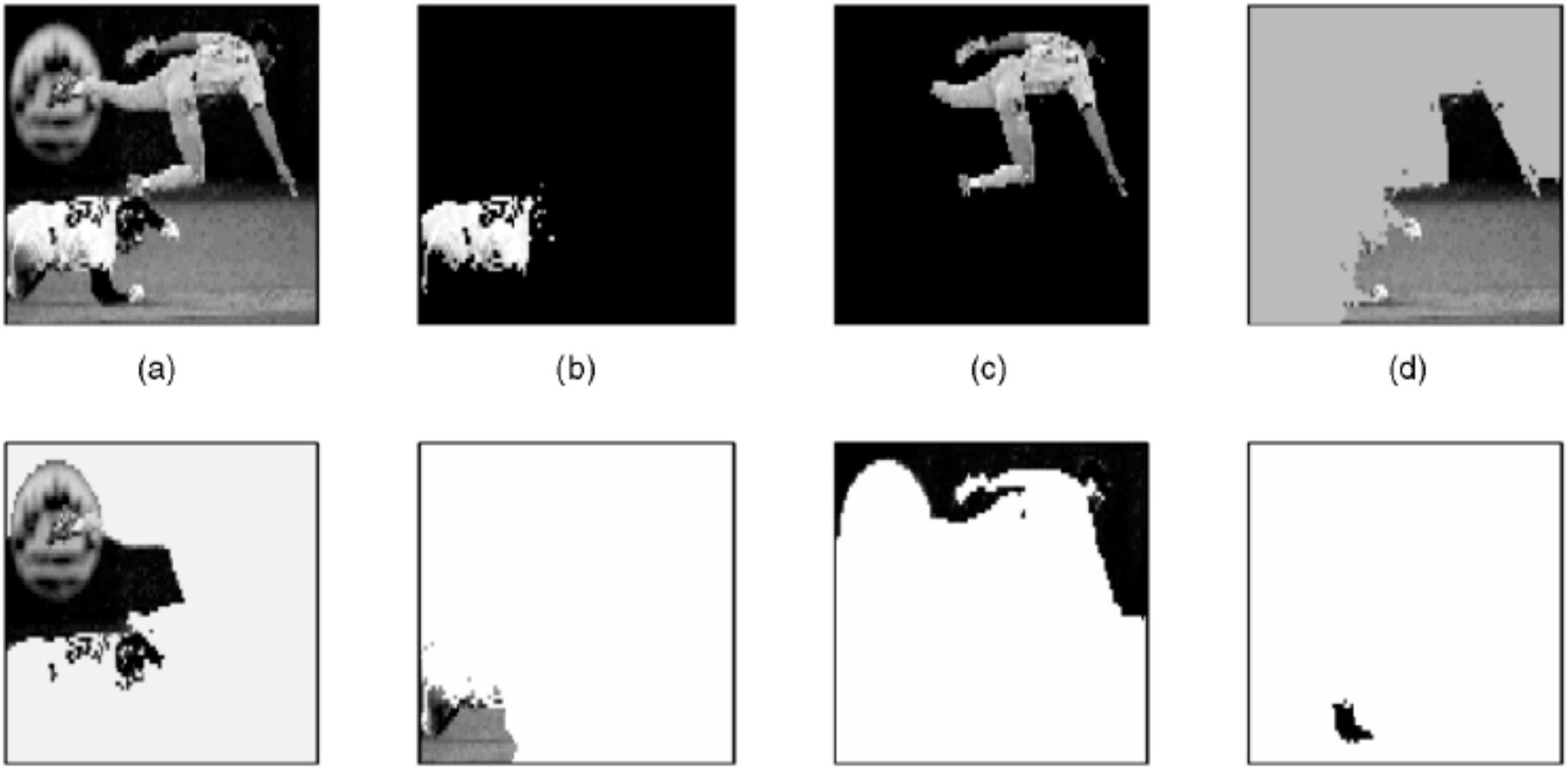
Graph theoretic approach to segmentation

Given an image for segmentation:

1. Set up a graph *G=(V,E)* with weights measuring similarities between pixels.
2. Compute *W,D,L*
3. Solve for the eigenvectors of $D^{-\frac{1}{2}}(\overbrace{D-W}^{L})D^{-\frac{1}{2}}$
4. Use the eigenvector of the second smaller eigenvalue to bipartition the graph.
5. Apply recursively as needed.

Segmentation Approaches

Graph theoretic approach to segmentation

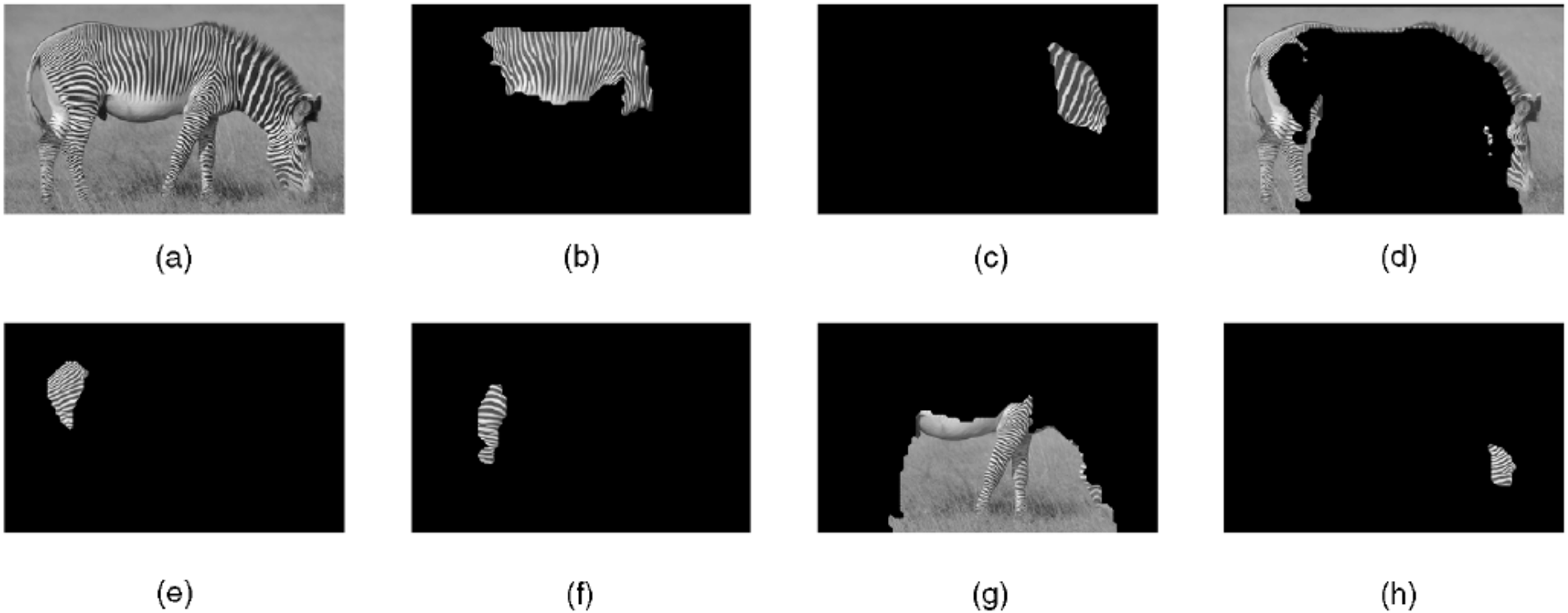


(a) (b) (c) (d)

(e) (f) (g) (h)

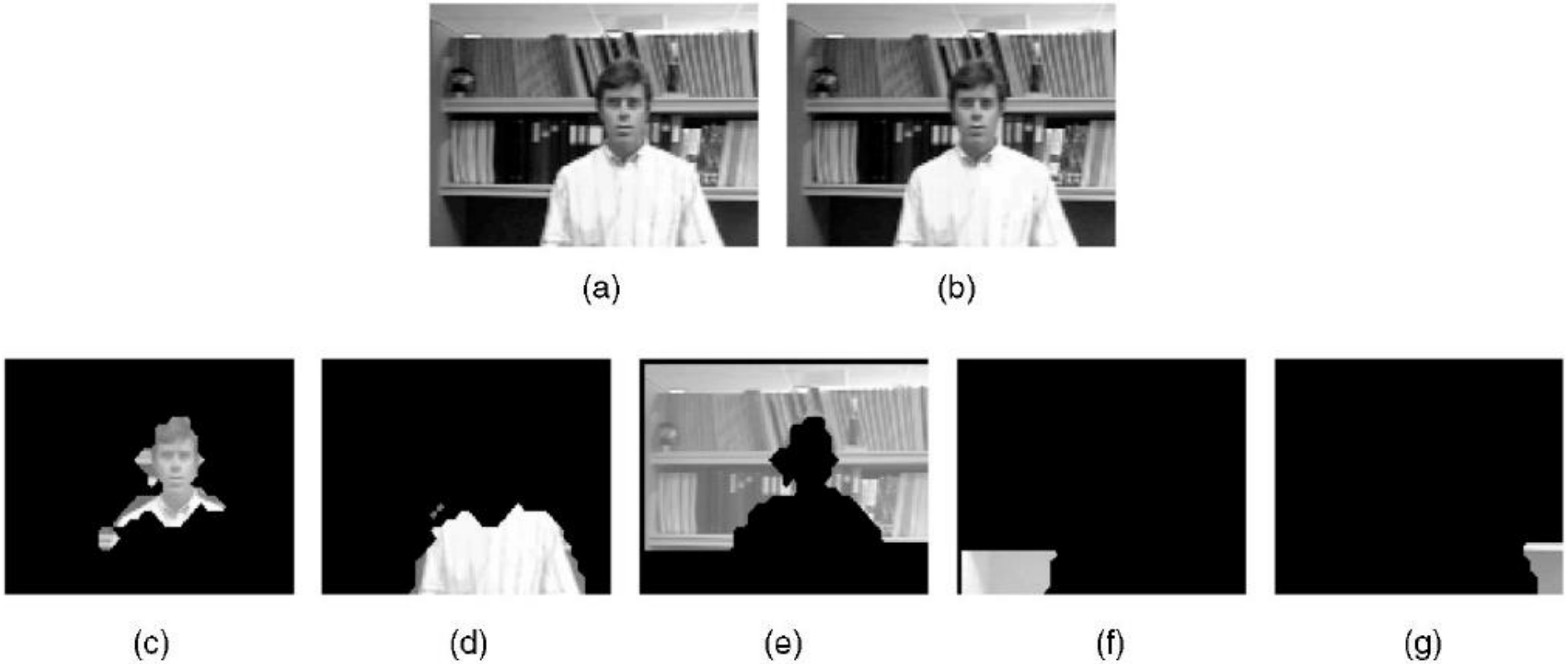
Segmentation Approaches

Graph theoretic approach to segmentation



Segmentation Approaches

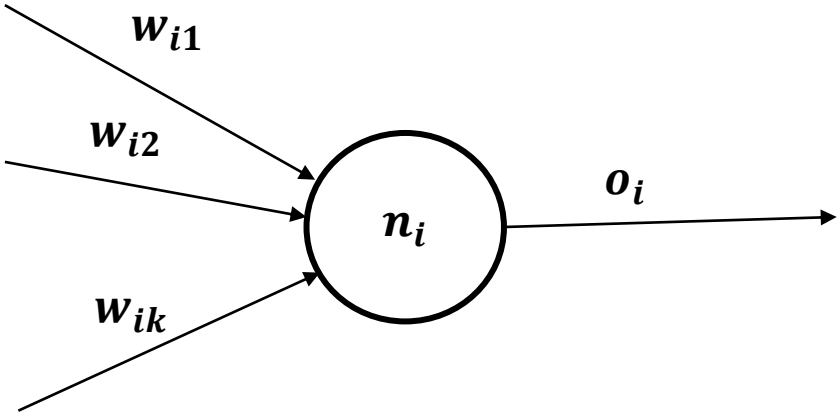
Graph theoretic approach to segmentation



Segmentation Approaches

Leap to the present - Machine learning and Semantic Segmentation

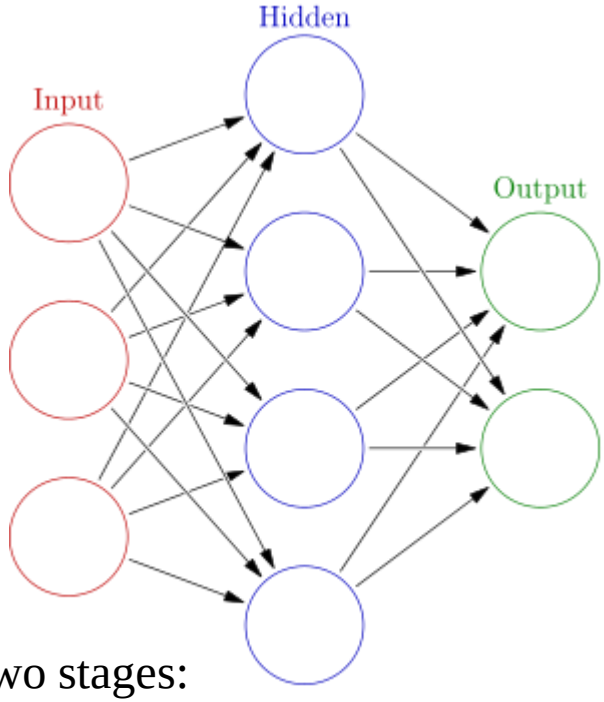
- Artificial neural networks (ANNs) are systems broadly inspired by the biological neural networks that constitute animal brains.
- ANNs consist of "neurons" where each neuron receives input signals, processes them (typically with a non linear activation function) and outputs a signal that drives consecutive neurons.
- Each edge has a weight that increases ("excites") or decreases ("inhibits") the unit activation based on the signal in that edge.
- It can be shown that it is possible to model any logical operator (e.g. AND, OR, XOR).



Segmentation Approaches

Leap to the present - Machine learning and Semantic Segmentation

- One popular class of ANNs are feedforward consist of an input layer and an output layer. In between, there can be any number of hidden layers.
- The structure of the network changes depending on the desired task.



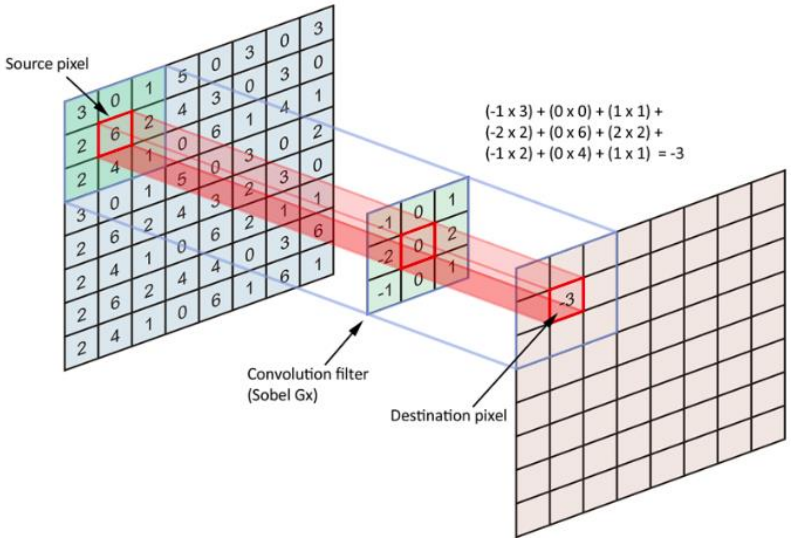
Working with ANNs is typically “supervised”, i.e., consists of two stages:

- **Training** - labeled data (e.g. pictures segmented by a human) is used together with a loss function to calculate optimal edge weights.
- **Classification -** the neural net is used to classify new input.

Segmentation Approaches

Leap to the present - Machine learning and Semantic Segmentation

- In computer vision, a commonly used network is a convolutional neural network (CNN), i.e., a layered network with weight sharing between different units in the hidden layer.
- The input layer's neurons can be the brightness or color values of an image's pixels.
- Each layer is a convolution of the previous layer.
- Kernel values are determined by edges weight and kernel size by network structure



Segmentation Approaches

Leap to the present - Machine learning and Semantic Segmentation

- A diverse dataset is segmented manually (can be millions of images)
- Networks designed by experts are used. They contain a combination of convolution layers and additional “non linear” layers.
- The training process can be very computational extensive. Advancement in GPU technologies made turned this process realistic in time.
- The results are phenomenal:

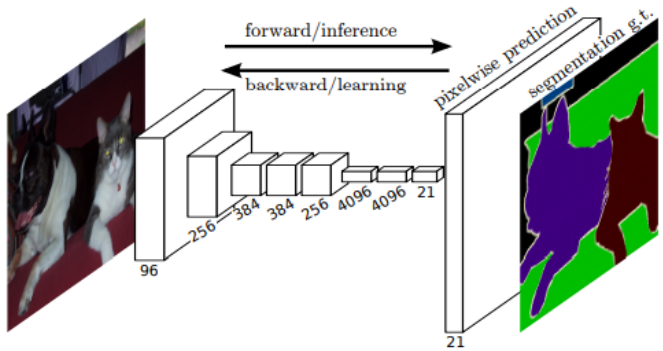
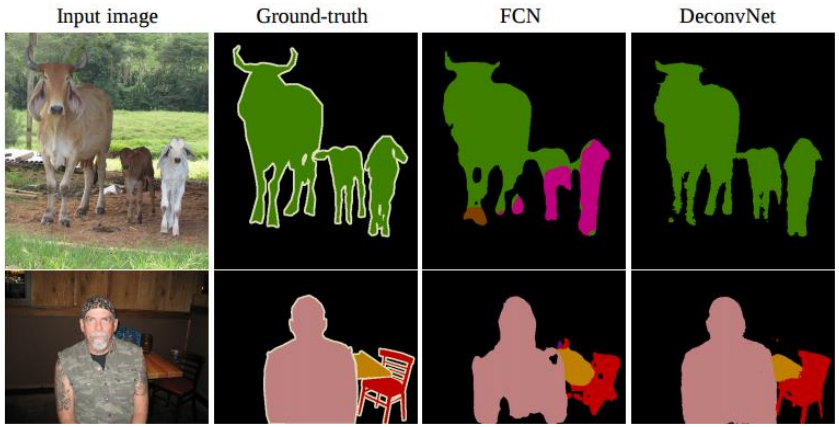
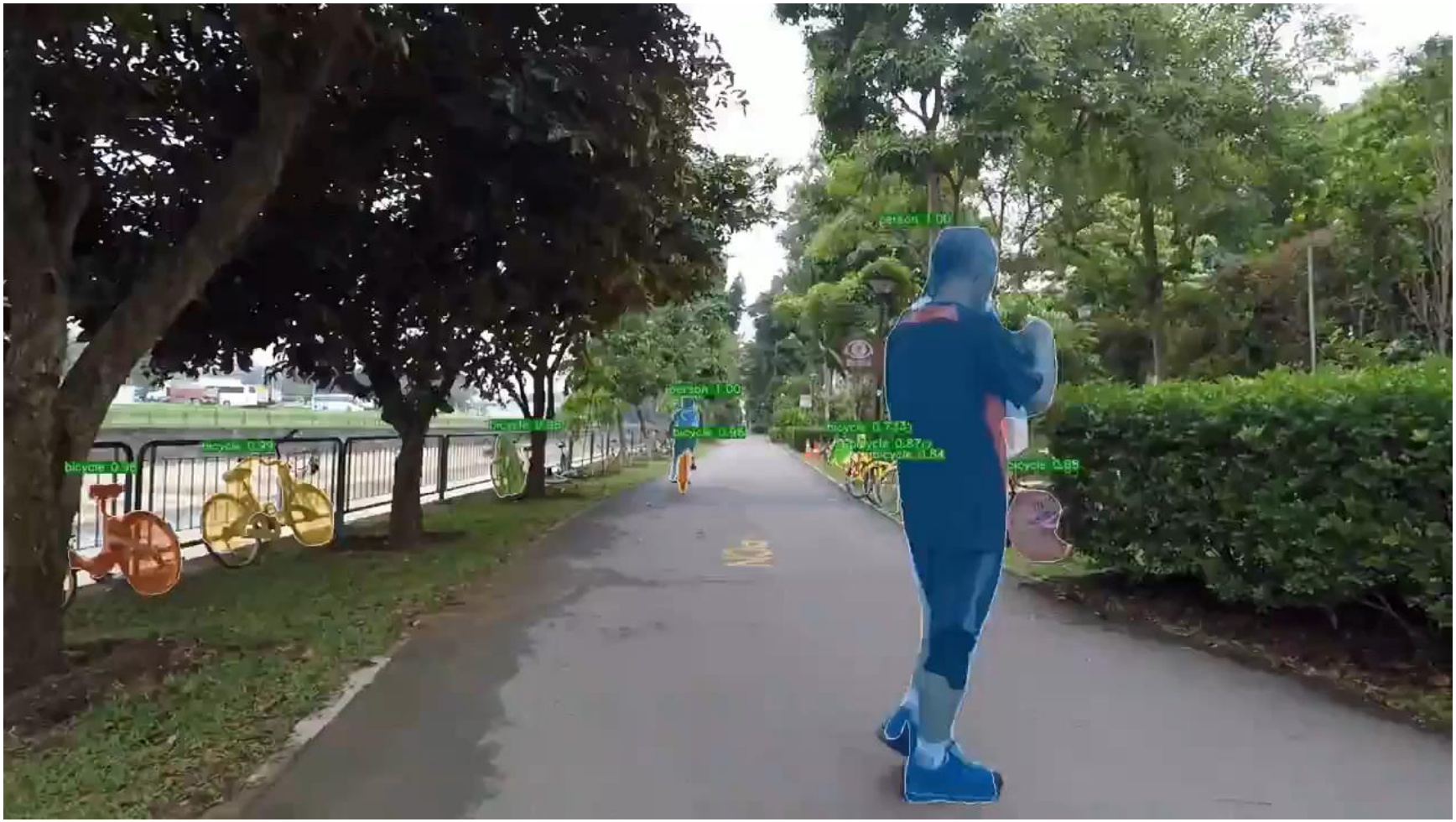


Figure 1. Fully convolutional networks can efficiently learn to make dense predictions for per-pixel tasks like semantic segmen-



Segmentation Approaches

Leap to the present - Machine learning and Semantic Segmentation